
THE SHAPE OF ATTRACTION IN UMAP: EXPLORING THE EMBEDDING FORCES IN DIMENSIONALITY REDUCTION

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ABSTRACT

Uniform manifold approximation and projection (UMAP) is one of the most widely used neighbor embedding methods. The method relies on attractive and repulsive forces among high-dimensional data points to obtain a low-dimensional embedding. In this paper, we dissect these forces and pinpoint exactly how they govern cluster formation and visualization, and we position UMAP precisely against its contemporaries. Repulsion emphasizes differences: it controls cluster boundaries and sets inter-cluster distance. Attraction is far more subtle, because the attractive tension between points manifests simultaneously as attraction and repulsion in the lower-dimensional mapping. This single mechanism explains why learning rate annealing is necessary and why attractive and repulsive terms demand different treatment. Building on this understanding, we modify attraction and thereby make cluster formation markedly more consistent under random initialization. In short, our analysis renders UMAP and similar embedding methods more interpretable, more robust, and more accurate.

1 Introduction

The current era is defined by a deluge of high-dimensional data. Dimensionality reduction (DR) techniques have become indispensable tools for exploratory analysis of such data, exposing the underlying structure through visualization. The two dominant methods, *t*-distributed stochastic neighbor embedding (Maaten and Hinton, 2008) and uniform manifold approximation and projection (UMAP) (McInnes et al., 2018), are both grounded in attraction-repulsion dynamics that pull similar data points together while pushing dissimilar ones apart. As unsupervised algorithms, they require no labeled data; instead, they recover and preserve the intrinsic structure of high-dimensional data by exploiting local (attractive) and global (repulsive) relationships (forces). This makes them the methods of choice for tasks such as clustering (Becht et al., 2019), exploratory data analysis (Fleischer and Islam, 2020), anomaly detection in semiconductor manufacturing (Fan et al., 2021), visual search (González-Márquez et al., 2024), time series analysis (Altın and Cakir, 2024), studying representation convergence (Huh et al., 2024), and outlier image detection (Islam and Fleischer, 2024), precisely where visualizing hidden patterns in unlabeled data is critical and meaningful. By learning embeddings in a data-driven, label-free manner, DR exemplifies the power of unsupervised methods to distill complex data into readily interpretable forms.

Building on the attraction-repulsion principle, a wave of newer methods has emerged (Amid and Warmuth, 2019; Agrawal et al., 2021; Wang et al., 2021; Narayan et al., 2021; Yang et al., 2022; Wang et al., 2025; Kury et al., 2025), each engineered to emphasize specific aspects of the data. Yet despite their reach across diverse applications, these methods rest on heuristic practices that routinely fail to yield meaningful interpretations. Moreover, DR introduces distortions that are unavoidable (Chari and Pachter, 2023). A deeper understanding of the algorithms is therefore essential, so that practitioners can interpret embeddings correctly, reject spurious structures, and optimize performance. In practice, these algorithms coax compact clusters into existence through an assortment of techniques—carefully chosen initialization, learning rate schedules, and kernel function tuning—yet the underlying dynamics of the attractive and repulsive forces that actually drive cluster formation have never been investigated in depth. Worse, the essential tunable parameters remain buried inside abstract functional forms, which obscures rather than explains what the algorithms do.

In this paper, we decompose the forces into their constituent parts and extract the functional shapes of attraction and repulsion. We demonstrate that three seemingly unrelated phenomena—the necessity of learning rate annealing, the

difficulty of producing consistent output under random initialization, and the very origin of cluster formation—all trace back to attraction. Repulsive forces, by contrast, govern inter-cluster distances. Our specific contributions are:

1. We formulate attraction and repulsion shapes directly from the attractive and repulsive forces, establish exact conditions for contraction and expansion of distance, and thereby open a new and clarifying perspective on these algorithms (Section 4).
2. We prove that the attraction shape of UMAP produces the counterintuitive coexistence of contraction and expansion of distance. By comparing attraction shapes across algorithms, we explain precisely how attraction dictates the learning rate annealing scheme (Section 5).
3. We modify the attraction shape and, in doing so, substantially improve the consistency of embeddings under random initialization—direct evidence that the data encode a unique structure (Section 5.1).
4. Analyzing repulsion shapes, we establish that repulsion governs cluster compactness and regulates inter-cluster distance (Section 5.2).
5. We compare the attraction and repulsion shapes of UMAP, Parametric UMAP, and NEG- t -SNE, expose their exact similarities and distinctions, and characterize the stability of each algorithm (Section 6).

We center the main text on UMAP, now the de facto standard across many fields, and bring in other algorithms where relevant. Further details on the individual algorithms appear in the appendix. The code used in this research will be released upon acceptance.

2 Related Works

Modern iterative graph-based neighbor embedding algorithms originate with stochastic neighbor embedding (SNE) (Hinton and Roweis 2002) and its t -distribution extension (t -SNE) (Maaten and Hinton 2008). Both methods build a dense graph in which every point has a pairwise relation with all others, irrespective of similarity. The graph weights are then normalized to define a probability distribution. Concurrent methods, including locally linear embedding (Roweis and Saul 2000) and Laplacian Eigenmaps (LE) (Belkin and Niyogi 2002), instead used a k -nearest neighbor (k -NN) graph of pairwise interactions. These spectral methods rely on eigenvalue decomposition. Tang et al. (Tang et al. 2016) then folded the k -NN graph into the iterative approach and dropped normalization in the lower dimension. McInnes et al. (2018) pushed this further in UMAP, removing normalization altogether—in both high and low dimensions—and obtaining the embedding from pairwise interactions alone. Its optimization applies an explicit attractive force to preserve the local neighborhood and a repulsive force to keep dissimilar points apart. Building on these foundations, methods such as PaCMAP (Wang et al. 2021) and NEG- t -SNE (Damrich et al. 2023) have followed. For a recent survey, see de Bodt et al. (2025).

Understanding the relationships among these algorithms has advanced considerably. An early analysis of SNE established that data well-clustered in the original space remain well-clustered in the embedding space (Shaham and Steinerberger 2017). A parallel analysis of t -SNE by Linderman and Steinerberger (2019) proved that the number of clusters in the embedding space lower-bounds the number in the original space. Subsequent work sharpened these characterizations (Arora et al. 2018; Cai and Ma 2022; Linderman and Steinerberger 2022). Because t -SNE and UMAP spring from the same underlying framework yet produce drastically different visualizations, a central thrust of the literature has been to pin down the connection between them (Böhm et al. 2022; Damrich et al. 2023; Draganov et al. 2023). Böhm et al. (2022) argued that Laplacian Eigenmap, ForceAtlas2 (Jacomy et al. 2014), UMAP, and t -SNE are all samples from one underlying spectrum. LE and t -SNE are, in fact, connected through the early exaggeration phase (Cai and Ma 2022), and UMAP and t -SNE are linked through contrastive estimation (Damrich et al. 2023). Around the same time, Hu et al. (2023) independently uncovered the relation between contrastive learning and SNE. More recent work explains dimensionality reduction from a probabilistic perspective (Ravuri et al. 2023; Ravuri and Lawrence 2024), through kernel techniques (Draganov and Dohn 2023), and via information geometry (Kolpakov and Rocke 2024).

3 Uniform Manifold Approximation and Projection

UMAP constructs a high-dimensional graph of the dataset $X = \{x_i \in \mathbb{R}^n | i = 1, \dots, N\}$ by using a pairwise relation: $p_{i,j} = f_h(d(x_i, x_j))$, typically, $\in [0, 1]$ (for details, see Appendix H).

The graph of the low-dimensional representation $Y = \{y_i \in \mathbb{R}^d | i = 1, \dots, N\}$ is given by a differentiable function

$$q_{ij} = \frac{1}{1 + a(\|y_i - y_j\|_2^2)^b}, \quad (1)$$

where the parameters a and b determine the density of the mapping and are chosen by fitting q_{ij} to

$$\Psi(d_{ij}) = \begin{cases} 1 & \text{if } \|y_i - y_j\|_2 < m_d \\ \exp(-(d_{ij} - m_d)) & \text{otherwise} \end{cases}, \quad (2)$$

where m_d regulates the distance between the two nearest low-dimensional points. UMAP aims to minimize the following cross-entropy loss function:

$$\mathcal{L} = \sum_{i,j} (-p_{ij} \log(q_{ij}) - (1 - p_{ij}) \log(1 - q_{ij})). \quad (3)$$

The first term supplies an attractive force and the second a repulsive force. Rather than optimizing every point at each iteration, UMAP adopts the negative sampling approach (Mikolov et al., 2013; Tang et al., 2016). For each edge with $p_{ij} > 0$ —a positive edge—it randomly samples several negative edges. The attractive force acts on the positive edge:

$$y_i^{t+1} = y_i^t + \lambda \nabla_{y_i^t} \log(q_{ij}), \quad (4)$$

$$y_j^{t+1} = y_j^t + \lambda \nabla_{y_j^t} \log(q_{ij}), \quad (5)$$

and the repulsive force acts on the negative edges:

$$y_i^{t+1} = y_i^t + \lambda \nabla_{y_i^t} \log(1 - q_{ij}), \quad (6)$$

where $\lambda(> 0)$ is the learning rate and t is the step number. Note that y_j is not updated for negative edges. For a detailed analysis of UMAP’s loss, see (Damrich and Hamprecht, 2021).

4 Attraction and Repulsion Shapes

The action of the updates (4)–(6) simplifies cleanly once we decompose the gradients $\nabla_{y_i^t} \log(q_{ij})$ ($\nabla_{y_i^t} \log(1 - q_{ij})$) into a scalar coefficient—dependent on the distance $\|y_i - y_j\|_2$ —acting on the vector $(y_i - y_j)$. We call this scalar coefficient the attraction (repulsion) shape. Although we use UMAP as the running example, the formalism applies to any method built on attraction and repulsion.

By writing $\nabla_{y_i^t} \log(q_{ij}) = f_a(\zeta^t)(y_i^t - y_j^t)$, where $\zeta = \|y_i - y_j\|_2$, we can update the equations of a positive edge as

$$y_i^{t+1} = y_i^t + \lambda f_a(\zeta^t)(y_i^t - y_j^t), \quad (7)$$

$$y_j^{t+1} = y_j^t - \lambda f_a(\zeta^t)(y_i^t - y_j^t). \quad (8)$$

Here, $f_a : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\leq 0}$ is the attraction shape, and we use the fact that for Euclidean metric, $\nabla_{y_i^t} \log(q_{ij}) = -\nabla_{y_j^t} \log(q_{ij})$. Similarly, we reformulate the update equation of a negative edge as

$$y_i^{t+1} = y_i^t + \lambda f_r(\zeta^t)(y_i^t - y_j^t), \quad (9)$$

$$y_j^{t+1} = y_j^t, \quad (10)$$

where $f_r : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ is the repulsion shape.

Such a decomposition has appeared before, e.g., in (McInnes et al., 2018; Agrawal et al., 2021; Draganov and Dohn, 2023), but earlier formulations put it to markedly different use. The original UMAP paper (McInnes et al., 2018) treated it merely as a computational trick for fast processing, and (Draganov and Dohn, 2023) used it to compare algorithms from a kernel perspective; in both cases the goal was simply computing the derivative. We instead treat the decompositions as independent functions free to take arbitrary forms—a shift in viewpoint that is the key to our analysis. (Agrawal et al., 2021) took a related step, but expressed the decomposition as a scalar coefficient times a unit vector $((y_i - y_j)/\|y_i - y_j\|_2)$ to emphasize force magnitude. As the discussion below makes clear, our shape decomposition is far more illuminating than magnitude alone.

4.1 Conditions for Attraction and Repulsion

In this section, we establish the exact conditions for attraction and repulsion directly from the update equations (7)–(10). The following proposition characterizes the contraction of distance between the pair y_i^t and y_j^t of a positive edge:

Proposition 4.1. *The update Eqs. (7) and (8) provide a contraction of distance ($\|y_i^{t+1} - y_j^{t+1}\| < \|y_i^t - y_j^t\|$) if $-1 < \lambda f_a < 0$.*

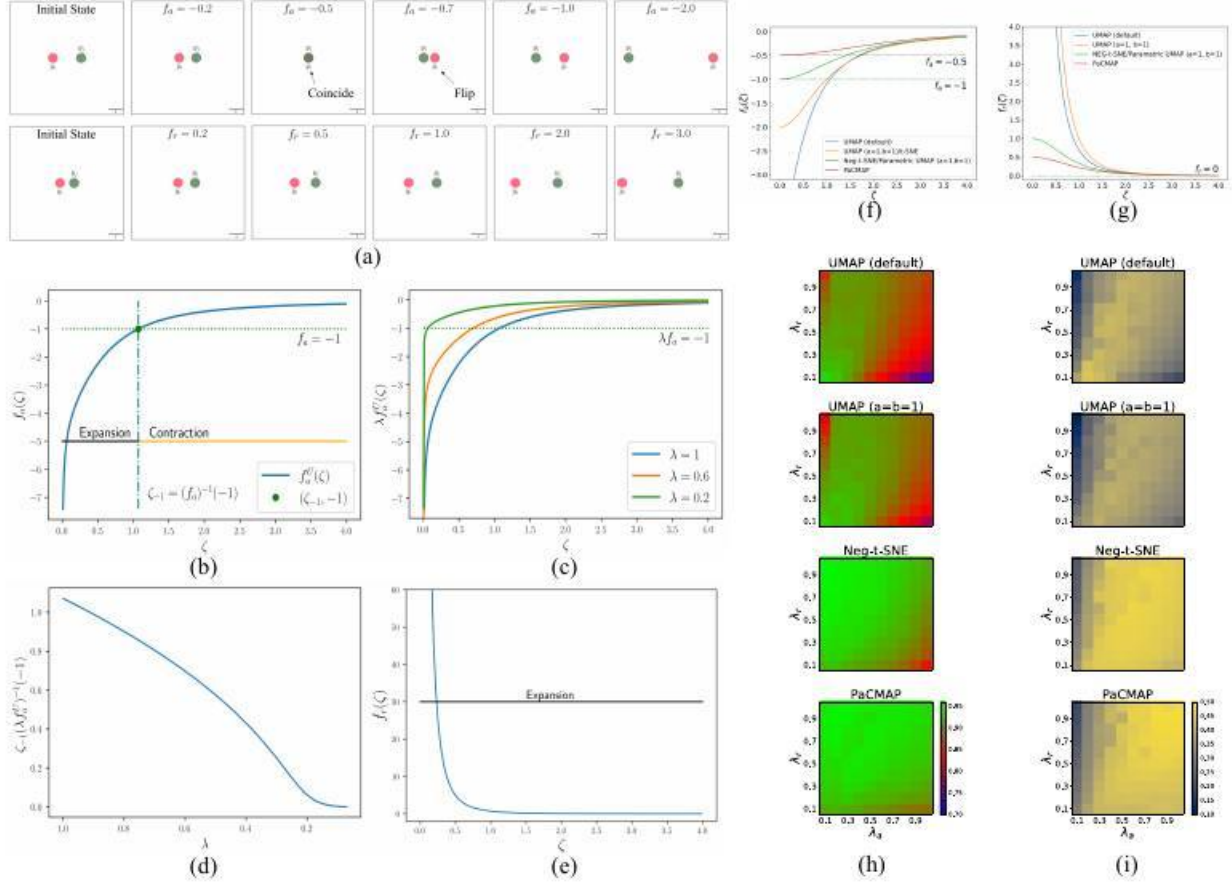


Figure 1: Attraction and repulsion shapes in UMAP. (a) Effect of different values of f_a (top) and f_r (bottom) on a pair. (b) Attraction shape of UMAP. (c) Effective attraction shape (λf_a) for various learning rates λ . (d) Minimum distance for contraction (ζ_{-1}) as λ decreases. (e) Repulsion shape of UMAP. (f) Attraction and (g) repulsion shapes of various embedding methods. (h) Trustworthiness and (i) Silhouette Score of various methods as the constant learning rate is varied for attraction (λ_a) and repulsion (λ_r) independently. Default UMAP parameters: $a = 1.58$ and $b = 0.89$.

Here λf_a acts as the effective attraction shape, though f_a alone already suffices to draw meaningful conclusions about the embeddings in nearly every case. For a negative edge, the following proposition characterizes the expansion of the distance between the pair y_i^t and y_j^t :

Proposition 4.2. *The update Eqs. (9) and (10) provide an expansion of distance ($\|y_i^{t+1} - y_j^{t+1}\| > \|y_i^t - y_j^t\|$) if $f_r > 0$.*

Note that including a symmetric term $-\lambda f_r(\zeta)(y_i^t - y_j^t)$ in Eq. (10) leaves this conclusion intact. Crucially, these conditions constitute a per-iteration certificate of guaranteed contraction/expansion; they must not be conflated with learning rate tuning or decay, because we can design the shapes so that contraction and expansion require no learning rate decay at all.

Figure 1(a) shows the effect of different values of $f_a < 0$ and $f_r > 0$ on two points. The repulsion shape behaves simply: any positive value increases the distance. The attraction shape is far subtler, because f_a encodes both attractive and repulsive dynamics at once. For $f_a \in (-1.0, 0)$ the distance decreases, with a sign flip at $f_a = -0.5$, the point of maximum attraction (coincident points). Any value below -1.0 instead increases the distance. Though these forces act purely locally, they collectively dictate the global structure.

5 Analysis of UMAP in terms of Attraction and Repulsion Shape

Using the gradient decomposition and the distance form [1](#), the attraction and repulsion shapes are given by

$$f_a^U(\zeta) = -\frac{2ab\zeta^{2(b-1)}}{1 + a\zeta^{2b}} \quad (11)$$

and

$$f_r^U(\zeta) = \frac{2b}{\zeta^{2b}(1 + a\zeta^{2b})}, \quad (12)$$

respectively. This section focuses on the default shapes of UMAP, controlled primarily by the parameters a and b , and extracts the insights that emerge from perturbing them. The discussion holds for $b \leq 1$, where the attraction shape is strictly increasing and hence invertible (for the derivation, see [Appendix A](#)).

Figure [1](#)(b) shows the default attraction shape of UMAP ($a = 1.58, b = 0.89$). It grows unbounded (approaches ∞) as $\zeta \rightarrow 0$. Exactly as [Proposition 4.1](#) predicts, the transition from contraction to expansion occurs when λf_a^U crosses -1 as ζ approaches 0. Because the attraction shape is invertible, we can identify the distance at this transition, $\zeta_{-1} = (\lambda f_a^U)^{-1}(-1)$, as the minimum distance for contraction under attractive updates. Thus $\zeta > \zeta_{-1}$ contracts and $0 < \zeta < \zeta_{-1}$ expands—flatly contradicting the intuition that attractive updates always bring points closer together.

When ζ_{-1} is high, neighboring points oscillate between contraction and expansion and the clusters turn fuzzy. The optimization goal for sharp boundaries therefore reduces to one clean requirement: drive $\zeta_{-1} \rightarrow 0$. This is precisely why UMAP’s learning rate schedule must anneal to zero (Figs. [1](#)(c,d)). The repulsion shape (Eq. [12](#)), by contrast, is always positive and satisfies [Proposition 4.2](#): f_r^U approaches 0 as $\zeta \rightarrow \infty$ and ∞ as $\zeta \rightarrow 0$ (Fig. [1](#)(e)).

Comparing attraction shapes across algorithms makes the picture unmistakable: only UMAP and t-SNE must contend with $f_a < -1$ (and hence $\zeta_{-1} > 0$ at $\lambda = 1$, Fig. [1](#)(f)). t-SNE handles this by weighting the updates with the corresponding p_{ij} values, whereas UMAP relies on learning rate annealing. Methods such as Neg-t-SNE and PaCMAP keep f_a naturally within $[-1, 0]$ (and therefore satisfy [Proposition 4.1](#) for $\lambda \in [0, 1]$ with $\zeta_{-1} = 0$). PaCMAP’s weighted f_a (< 0.5 throughout) goes further still, ruling out flips during attraction entirely. The repulsion shapes tell an equally sharp story: only UMAP must cope with large values at small distances (Fig. [1](#)(g)); the shape is unbounded, and the values are routinely clipped during optimization. For further details on these and related methods, see [Appendix G](#).

We demonstrate the impact of these attraction/repulsion choices on MNIST by applying separate constant learning rates for attraction (λ_a) and repulsion (λ_r) and sweeping their values (Figs. [1](#)(h,i)). For UMAP, a low λ_a (< 0.5 , so that $\zeta_{-1} \simeq 0$) yields the best embedding (trustworthiness (T) [Venna and Kaski, 2001](#), a measure of structure preservation, in Fig. [1](#)(h)) and clustering (silhouette scores (SIL) [Rousseeuw, 1987](#), a measure of cluster separation, in Fig. [1](#)(i)), while the entire range of λ_r ($\in [0, 1]$) remains usable. For NEG-t-SNE and PaCMAP, where $\zeta_{-1} = 0$, the full range of parameters works. This confirms unambiguously that UMAP depends on driving ζ_{-1} close to 0 for satisfactory embedding and clustering—which it accomplishes through learning rate annealing—whereas the newer algorithms have $\zeta_{-1} = 0$ built in by default. For further discussion, results, and illustrations across datasets, see [Section 6](#) and [Appendices B, G](#) and [I](#).

5.1 Improving UMAP’s Consistency under Random Initialization

The consistency of UMAP embeddings hinges on proper initialization [Kobak and Linderman, 2021; Wang et al., 2021](#). Typically, principal component analysis (PCA) of the data or spectral decomposition of the high-dimensional graph initializes the embedding, delivering consistent mappings despite various sources of stochasticity. Under random initialization, however, clusters frequently fail to form or emerge in a different orientation on every run. When the initial distance between two points (nearest neighbors in high dimension) is large, the attractive forces are simply too weak to pull them together. This phenomenon—near-sightedness [Wang et al., 2021](#)—is written directly into the attraction shape, where $|f_a^U|$ decays to zero as the distance grows (i.e., $\lim_{\zeta \rightarrow \infty} |f_a^U(\zeta)|\zeta = 0$).

The remedy follows immediately: induce “far-sightedness” by boosting attraction at large distances so that faraway neighbors are drawn together. To test this, we modify UMAP’s attraction shape to strengthen the attractive force:

$$f_a^M = f_a^U - \beta\zeta, \quad (13)$$

where β regulates the strength of the added term (we used $\beta = 0.2$, Fig. [2](#)(e)). This addition to the attraction shape corresponds exactly to a regularizer on the attractive term of the loss function (i.e., $\mathcal{L}^M = \mathcal{L} + \sum_{i,j} \beta/3 \|y_i - y_j\|_2^3$). Beyond attracting distant pairs, this technique promotes the intermixing of points that drives convergence under random

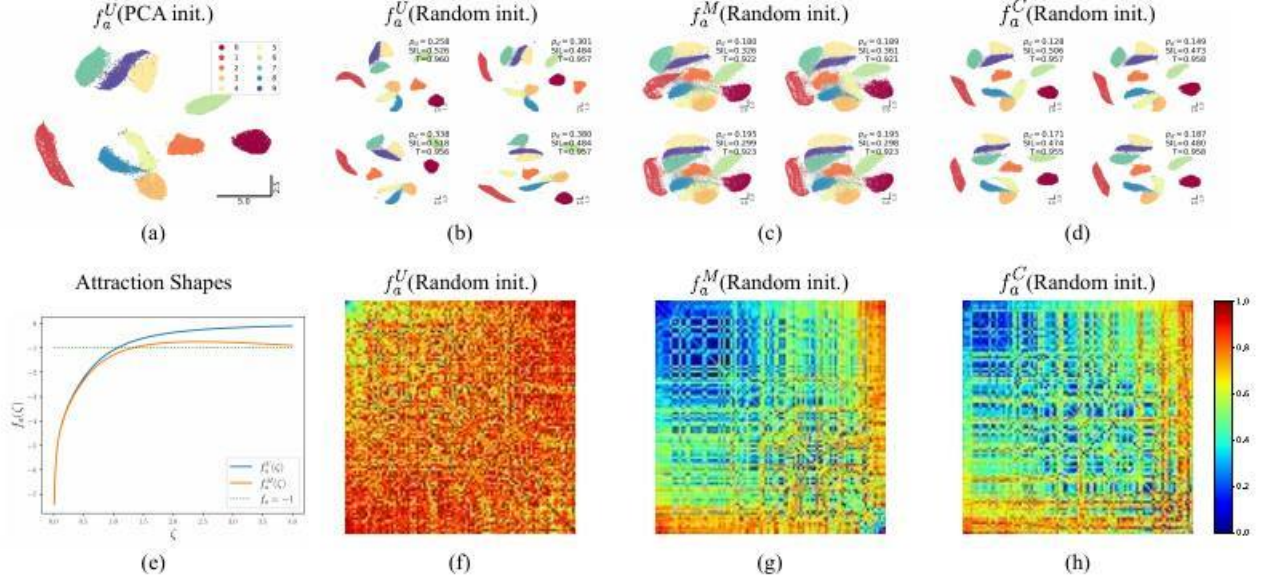


Figure 2: Effect of random initialization across attraction shapes on the MNIST dataset. (a) PCA mapping. (b-d) The four mappings with the lowest Procrustes distance (p_d) from the embedding in (a) for (b) UMAP, (c) modified, and (d) composite attraction shapes. (e) Default UMAP and modified attraction shapes. (f-h) Procrustes matrices from 100 runs of (f) UMAP (0.78 ± 0.13), (g) modified (0.49 ± 0.21), and (h) composite attraction (0.50 ± 0.20) shapes. The diagonal (i, i) entries of the Procrustes matrix are sorted by Procrustes distance (p_d) from (a), and the off-diagonal (i, j) entries give p_d between the i^{th} and j^{th} mappings. The matrices and (mean $p_d \pm \text{std}$) values establish that UMAP’s embeddings are not self-similar, whereas the modified and composite attraction shapes clearly enforce scale-invariant structure.

initialization (akin to early exaggeration in t -SNE). In [13] we deliberately chose the simplest linear correction; other functions, such as $\log \zeta$ or ζ^p ($p \in \mathbb{R}_{\geq 0}$), work equally well.

We also consider a composite attraction shape:

$$f_a^C = \begin{cases} f_a^M, & \text{epoch} \leq 100 \\ f_a^U, & \text{otherwise} \end{cases}. \quad (14)$$

The composite shape removes any distortions introduced by f_a^M by reverting to the original UMAP once intermixing has done its work. Below, we quantify the effects of both the modified and composite attraction shapes on DR from random initialization.

We first built a PCA-initialized embedding of the MNIST dataset (Figure 2(a)). We then produced embeddings from random (Gaussian) initialization for each shape, repeating the experiment 100 times. To quantify the results, we use Procrustes analysis [Gower, 1975], which aligns two point clouds under scaling, translation, rotation, and reflection (for details, see Appendix D.1). We align the randomly initialized embeddings to the PCA-initialized one and measure their separation via the Procrustes distance (p_d). Figure 2(b) shows the four embeddings with the lowest p_d : the cluster shapes are consistent, but their placements are not. The modified and composite attraction shapes (Figs. 2(c) and (d), respectively) deliver markedly more consistent cluster placements.

To quantify placement further, we use the Procrustes matrix: its diagonal is sorted by p_d from the PCA-initialized mapping, and its off-diagonal values give p_d between two randomly initialized mappings (for details, see Appendix D.1). This quantification parallels the similarity matrix [Foote, 1999]. Embeddings from the default UMAP attraction shape bear little resemblance to one another (Fig. 2(f)), whereas the modified (Fig. 2(g)) and composite (Fig. 2(h)) shapes are strongly self-similar. This is direct evidence that UMAP encodes a unique structure independent of the initialization; the default algorithm simply fails to reach it because attraction at longer distances is too weak.

5.2 Cluster Formation and Compactness

The primary user-facing parameter for cluster formation in UMAP is the minimum distance m_d (via Eq. 2). Varying m_d , however, tightly constrains the accessible values of a and b (Fig. 3(a)). Consequently, reducing m_d typically produces embeddings with no additional benefit (Figs. 3(b-d)). The reason is clear: varying m_d barely moves the repulsion shape

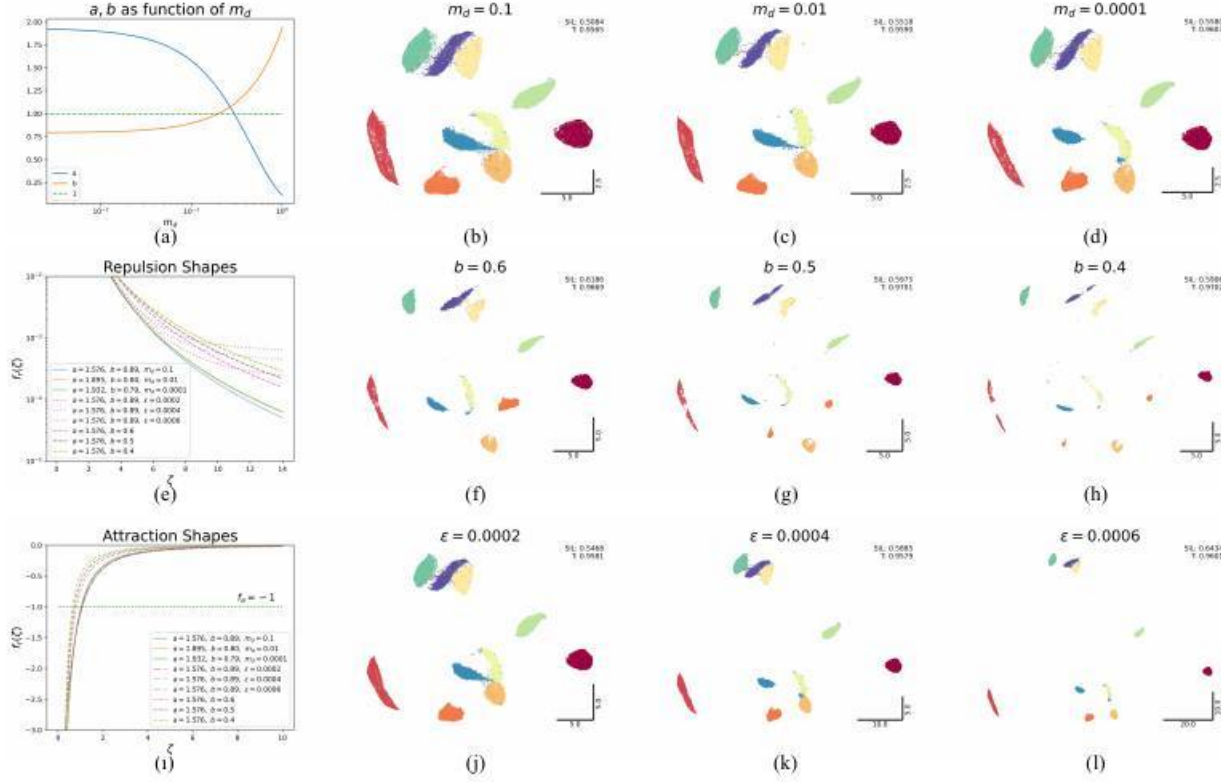


Figure 3: Control of inter-cluster distances on the MNIST dataset. (a) Computing a, b by varying the low-dimensional distance m_d restricts exploration. (b-d) UMAP output with m_d set to 0.1, 0.01, and 0.0001, respectively, shows negligible improvement in cluster compactness. (e) Repulsion shapes for different parameters. (f-h) Increasing repulsion by explicitly varying b yields more compact clusters and forms new ones that were otherwise absent. (i) Attraction shapes for varying parameters. (j-l) Increasing repulsion by adding a small positive value (ϵ) to the repulsion shape enlarges inter-cluster distance.

(Fig. 3(e)). We can instead vary a and b directly. Decreasing a increases repulsion but suppresses attraction even faster, aggravating near-sightedness. Decreasing b affords far better control (Figs. 3(e,i)). Figures 3(f-h) show inter-cluster distance growing and existing clusters splitting as b decreases to 0.6, 0.5, and 0.4, respectively. This splitting stems from the increasingly heavy-tailed kernel as b shrinks (heavy-tailed kernels yield smaller, distinct clusters [Van der Maaten and Hinton, 2008, Kobak et al., 2019]; see [Kobak et al., 2019] for how varying b modulates UMAP’s tail). Although this cleanly separates all ten MNIST labels, it conflates the two forces: lowering b amplifies repulsion more than lowering m_d does, yet it simultaneously reshapes the attraction profile.

To quantify the effect of repulsion independently, we can keep attraction fixed and vary the other. To achieve this, we modify the repulsion shape by adding a small positive value (ϵ):

$$f_r^M = f_r^U + \epsilon, \quad (15)$$

while holding a and b fixed, which is equivalent to adding a regularizer to the repulsive term of the loss function (i.e. $\mathcal{L}^M = \mathcal{L} - \sum_{i,j} \epsilon/2 \|y_i - y_j\|_2^2$). This gives us stronger repulsion than was previously attainable (Fig. 3(e)). Figs. 3(j-l) show inter-cluster distances growing steadily as ϵ increases. The clustering behavior, however, mirrors that obtained by varying m_d , yielding only a loose separation of all labels as ϵ grows. Critically, ϵ leaves the attraction shape untouched, and varying m_d traces essentially the same attraction shapes (Fig. 3(i))—strong evidence that cluster formation is governed predominantly by attraction.

To settle the question of cluster formation, we alter the attraction shape or the repulsion shape individually—leaving the other at default—by simply setting b to 0.4 (Fig. 4). The default attraction produces no new structures or clusters, while the increased repulsion ($b = 0.4$) merely yields smaller clusters than original UMAP (Fig. 4(a)). When attraction is increased instead—setting $b = 0.4$ with the default repulsion—the embedding reveals additional structures within each cluster (Fig. 4(b)), and some older clusters split into smaller ones. This establishes the point decisively: attraction drives cluster formation, while repulsion controls compactness (either by forming smaller clusters or by increasing

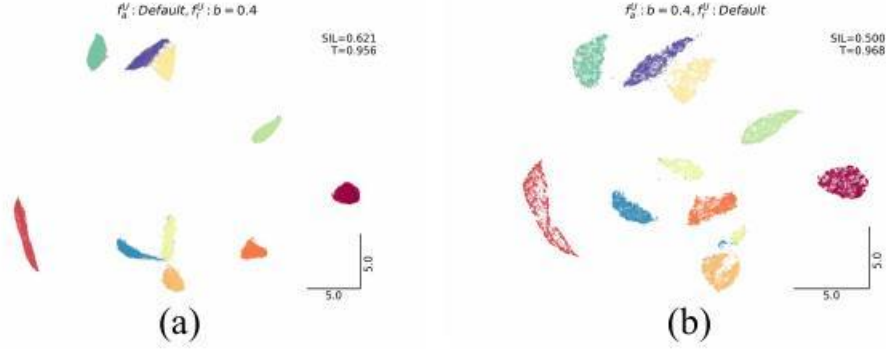


Figure 4: Embedding of the MNIST dataset with (a) default attraction shape but repulsion shape at $b = 0.4$ and (b) default repulsion shape but attraction shape at $b = 0.4$. The former reproduces the same clusters as default UMAP, only more compact. The latter develops new structures within clusters and forms entirely new clusters.

inter-cluster distance, depending on the repulsion shape). In Appendix G.3 we show that LocalMAP embeddings exploit this very interplay of attraction and repulsion to separate clusters.

6 Comparison to NEG- t -SNE

Setting a and b to 1 is standard across many dimensionality reduction algorithms, because it renders their gradients stable (bounded) and thereby eases optimization. Damrich et al. (2023) recently pursued this for UMAP and introduced Neg- t -SNE, which not only stabilizes the gradient but also produces more compact clustering even at a constant learning rate of 1. Parametric UMAP (Sainburg et al. 2021), meanwhile, began with the UMAP loss but later adopted a numerically stable modified cross-entropy loss (Shi et al. 2023) with a logsigmoid kernel. Analyzing both reveals that these two approaches converge on the same formulation from different starting points, which yields the following result:

Proposition 6.1. *Neg- t -SNE is Parametric UMAP with $a = 1$ and $b = 1$.*

With $a = 1$ and $b = 1$, the attraction and repulsion shapes of UMAP are given by $f_a^U = -2/(1 + \zeta^2)$ and $f_r^U = 2/(\zeta^2(1 + \zeta^2))$, respectively. The attraction shape is now bounded within $[-2, 0]$, with $f_a^U(0) = -2$ (Fig. 1(f)), while the repulsion shape is essentially unchanged (unbounded as $\zeta \rightarrow 0$, Fig. 1(g)). Because $f_a^U < -1$ as $\zeta \rightarrow 0$, Proposition 4.1 and the argument of Section 5 guarantee that even this unity case still demands learning rate annealing.

Damrich et al. (2023) compared UMAP’s negative sampling loss through the lens of contrastive embedding (CE) and concluded that UMAP’s effective kernel is $1/\zeta^2$. Within the CE framework, the authors defined NEG- t -SNE by switching the kernel to $1/(1 + \zeta^2)$. Translated back to the UMAP formalism, this gives the low-dimensional affinity $q_{ij}^N = 1/(2 + \zeta^2)$, and hence the attraction and repulsion shapes $f_a^N = -2/(2 + \zeta^2)$ and $f_r^N = 2/((1 + \zeta^2)(2 + \zeta^2))$, respectively (Figs. 1(f,g)). The attraction shape is now bounded within $[-1, 0]$ and satisfies Proposition 4.1. Any $\lambda \in [0, 1]$ therefore contracts and never oscillates between expansion and contraction. NEG- t -SNE is thus far less sensitive to learning rate annealing, and its clusters stay crisp even at constant $\lambda = 1.0$. Its repulsion shape is likewise bounded within $[0, 1]$ and never diverges as $\zeta \rightarrow 0$, so far fewer points escape the clusters when sampled randomly. Whereas Damrich et al. (2023) attributed this disparity solely to NEG- t -SNE’s bounded and UMAP’s unbounded repulsive forces, our analysis in Fig. 1 demonstrates that the attraction shape is equally responsible for the stability and compactness of the clusters. For further discussion and illustration of NEG- t -SNE, see Appendix F. Combined with the results in Fig. 4, this shows that attraction and repulsion shapes from different methods can be freely mixed—for example, pairing UMAP’s attraction shape with NEG- t -SNE’s repulsion shape. We give several such examples on UMAP, NEG- t -SNE, and PaCMAP in Appendix J.

7 Discussion and Conclusion

In this work, we pinned down how attractive and repulsive forces produce cluster formation. Although it is well known that attractive forces pull similar points together and repulsive forces push dissimilar points apart, the exact mechanisms behind these forces had never been properly characterized. Here, we demystified the dynamics underlying

cluster formation. While the main text centers on UMAP and NEG- t -SNE, our results generalize across dimensionality reduction (further algorithms appear in Appendix G).

We treated the attraction and repulsion coefficients as functional mappings and analyzed them through their respective shapes. Characterizing these shapes (Propositions 4.1 and 4.2) exposed a striking, counterintuitive fact: UMAP’s attractive forces actually repel (expand distance) at short distances (Fig. 1(b))—instead of drawing neighboring points together, they push them apart. UMAP’s learning rate schedule exists precisely to counteract this effect.

We also examined the initialization of the low-dimensional embedding. Similar points that start far apart experience too little attraction and never contract sufficiently. Our formalism offers a direct lever on attraction at larger distances through additional terms in the attraction shape. This modification produced outputs that are substantially more consistent under random initialization—firm evidence that UMAP encodes a unique structure, with numerical tricks serving only to reach it faster.

Analyzing the repulsion shape established that higher repulsion produces larger inter-cluster distances. Exploiting this, we added a small positive value to the shape to obtain repulsion stronger than any attainable by the conventional route of varying the minimum distance m_d .

Finally, we settled the relation between NEG- t -SNE and UMAP. In the case $a = 1, b = 1$, the decisive difference is that NEG- t -SNE, in its default form, has bounded attraction ($\in [-1, 0]$) and repulsion ($\in [0, 1]$) shapes, so both propositions hold regardless of the learning rate. The method is therefore far less sensitive to a fixed learning rate ($\in [0, 1]$). Varying a and b in NEG- t -SNE reintroduces exactly the numerical challenges of UMAP. We further proved that NEG- t -SNE and Parametric UMAP are equivalent.

These insights into attraction–repulsion dynamics provide concrete new tools for optimizing dimensionality reduction algorithms. Moreover, the tight connection between dimensionality reduction and contrastive learning (Damrich and Hamprecht 2021, Hu et al. 2023) indicates that our approach will also strengthen representation learning. Taken together, our work makes embeddings and their interpretations more principled, more consistent, and more reliable, and charts a clear path for future research.

Software and Data

All data used in this paper are publicly available. The MNIST dataset is available at yann.lecun.com. Fashion-MNIST is available at github.com. Single-cell transcriptomes data is available at github.com.

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Appendix

Appendix [A](#) supplies the full derivations of UMAP’s attraction and repulsion shapes. Building on the main text, Appendix [B](#) revisits the constant learning rate. Appendix [C](#) shows what improved attraction and repulsion deliver for large language model embeddings. Appendix [D.1](#) formally defines the Procrustes distance and matrix. Appendix [E](#) extends the random-initialization experiment to further datasets, and Appendix [F](#) extends the NEG- t -SNE and UMAP analysis. Appendix [G](#) works through alternate dimensionality reduction methods, and Appendix [H](#) details, for completeness, how these methods construct the high-dimensional graph. Appendix [I](#) gives the detailed results of varying λ_a and λ_r (from Fig. [1](#)) for UMAP, NEG- t -SNE, and PaCMAP across datasets. Appendix [J](#) demonstrates the effect of mixing and matching shapes from different algorithms. Finally, Appendix [K](#) provides implementation details.

A Proofs and Derivations

A.1 Proof of Proposition [4.1](#)

Proof. We subtract Eq. [\(7\)](#) from Eq. [\(8\)](#) and take a norm:

$$\|y_i^{t+1} - y_j^{t+1}\| = |1 + 2\lambda f_a| \|y_i^t - y_j^t\|. \quad (16)$$

This distance contracts as long as $|1 + 2\lambda f_a| < 1$, i.e., provided

$$-1 < \lambda f_a < 0, \quad (17)$$

□

A.2 Proof of Proposition [4.2](#)

Proof. We subtract Eq. [\(9\)](#) from Eq. [\(10\)](#) and take a norm:

$$\|y_i^{t+1} - y_j^{t+1}\| = |1 + \lambda f_r| \|y_i^t - y_j^t\|. \quad (18)$$

This distance increases when $|1 + \lambda f_r| > 1$, i.e.,

$$\lambda f_r < -2 \text{ or } f_r > 0. \quad (19)$$

From the definition of f_r , the latter suffices. □

A.3 Attraction Shape:

We use a general form of the low-dimensional affinity function, i.e., $q_{ij} = (\gamma + a\|y_i - y_j\|_2^{2b})^{-1}$, to derive the attraction shape. It reduces to UMAP for $\gamma = 1$ and to NEG- t -SNE for $\gamma = 2$. The attractive force is given by

$$\begin{aligned} \nabla_{y_i} \log q_{ij} &= -\nabla_{y_i} \log (\gamma + a(\|y_i - y_j\|_2^2)^b) \\ &= -\frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} \nabla_{y_i} (\gamma + a(\|y_i - y_j\|_2^2)^b) \\ &= -\frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} ab(\|y_i - y_j\|_2^2)^{b-1} \nabla_{y_i} \|y_i - y_j\|_2^2 \\ &= -\frac{2ab(\|y_i - y_j\|_2^2)^{b-1}}{\gamma + a(\|y_i - y_j\|_2^2)^b} (y_i - y_j). \end{aligned} \quad (20)$$

Defining $\zeta = \|y_i - y_j\|_2$, the first term gives the attraction shape as:

$$f_a(\zeta) = -\frac{2ab\zeta^{2(b-1)}}{\gamma + a\zeta^{2b}} \quad (21)$$

A.4 Condition for strictly increasing f_a^U :

Its behavior with distance can be characterized by computing the derivative of f_a^U :

$$\frac{df_a^U(\zeta)}{d\zeta} = -\frac{2ab\zeta^{2b-3}}{\gamma + a\zeta^{2b}} \left((b-1) - \frac{ab\zeta^{2b}}{\gamma + a\zeta^{2b}} \right). \quad (22)$$

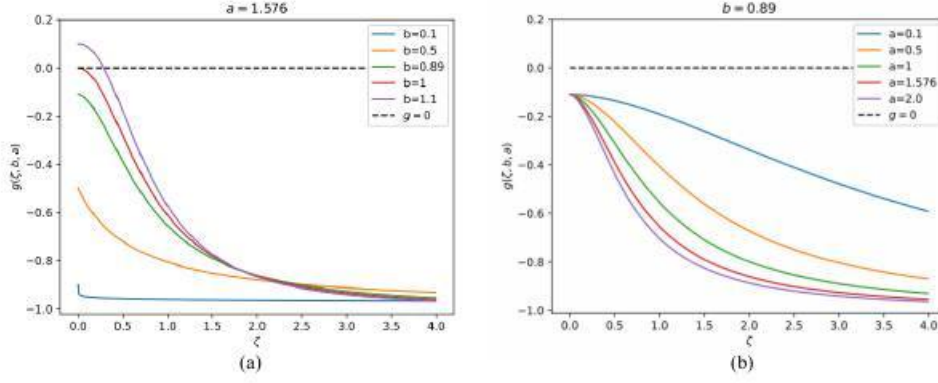


Figure 5: Values of $g(\zeta, b, a)$ for (a) a fixed at 1.576 and (b) b fixed at 0.89.

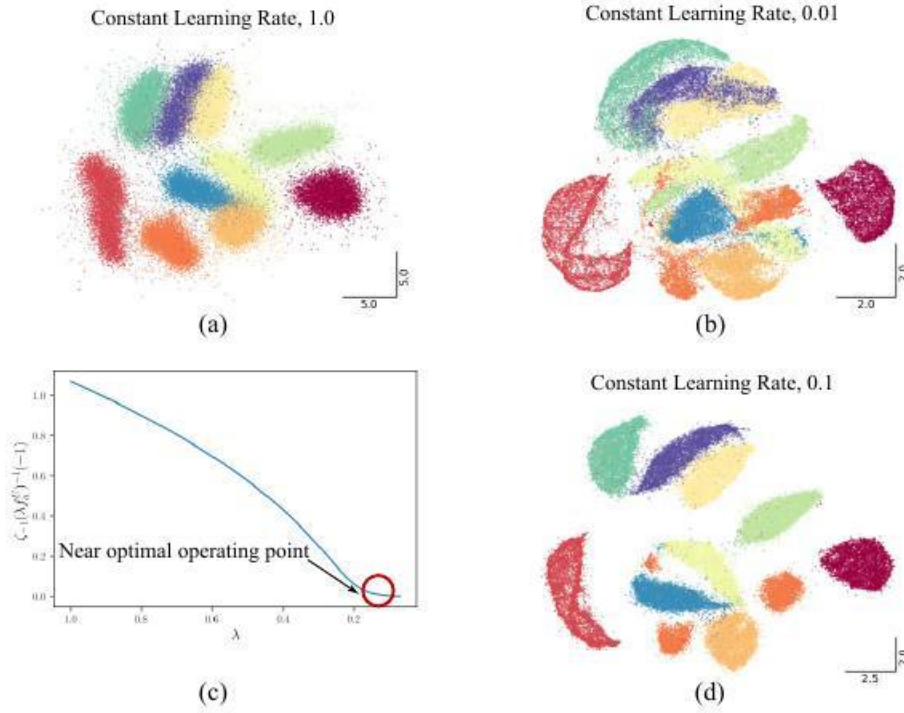


Figure 6: Effect of a constant learning rate on embeddings. (a) When the learning rate is too high ($\lambda = 1.0$), the embeddings are diffuse (owing to the large value of ζ_{-1}). (b) When the learning rate is too low ($\lambda = 0.01$), clusters fail to form (attraction and repulsion are both too weak). (c) ζ_{-1} decreases nonlinearly as the learning rate decreases; the algorithm's goal is to reduce ζ_{-1} while retaining effective levels of attraction and repulsion. (d) Distinct, compact clusters form at the constant, near-optimal learning rate $\lambda = 0.1$.

This leads to a strictly increasing condition ($\frac{df^U}{d\zeta} > 0$),

$$g(\zeta, b, a) < 0, \quad (23)$$

where $g(\zeta, b, a) = b - 1 - \frac{ab\zeta^{2b}}{\gamma + a\zeta^{2b}}$. This inequality is valid as long as $0 < b \leq 1$ (using the derivative and asymptotes of g). Figure 5 shows values of g for different b and a . As b increases above 1, the inequality (23) no longer holds.

A.5 Repulsion Shape:

The repulsive force, using the general form of the low-dimensional affinity, is given by

$$\begin{aligned}
 \nabla_{y_i} \log(1 - q_{ij}) &= \nabla_{y_i} \log \left[1 - \frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} \right] \\
 &= \frac{\gamma + a(\|y_i - y_j\|_2^2)^b}{(\gamma - 1) + a(\|y_i - y_j\|_2^2)^b} \nabla_{y_i} \left[1 - \frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} \right] \\
 &= \frac{1}{\gamma - 1 + a(\|y_i - y_j\|_2^2)^b} \frac{ab(\|y_i - y_j\|_2^2)^{b-1}}{\gamma + a(\|y_i - y_j\|_2^2)^b} \nabla_{y_i} \|y_i - y_j\|_2^2 \\
 &= \frac{2ab(\|y_i - y_j\|_2^2)^{b-1}}{(\gamma - 1 + a(\|y_i - y_j\|_2^2)^b)(\gamma + a(\|y_i - y_j\|_2^2)^b)} (y_i - y_j).
 \end{aligned} \tag{24}$$

The first term gives the repulsion shape as:

$$f_r(\zeta = \|y_i - y_j\|_2) = \frac{2ab\zeta^{2(b-1)}}{(\gamma - 1 + a\zeta^{2b})(\gamma + a\zeta^{2b})}. \tag{25}$$

Generally, $f_r > 0$.

A.6 Loss functions due to modified attraction and repulsion shape:

The cost function of the attractive term with the modification in Eq. (13) is given by

$$-\int (f_a^U(\|y_i - y_j\|_2) - \beta\|y_i - y_j\|_2)(y_i - y_j)dy_i = -\log(q_{ij}) + \frac{\beta}{3}\|y_i - y_j\|_2^3, \tag{26}$$

whereas the repulsive term due to Eq. (15) is given by

$$-\int (f_r^U(\|y_i - y_j\|_2) + \epsilon)(y_i - y_j)dy_i = -\log(1 - q_{ij}) - \frac{\epsilon}{2}\|y_i - y_j\|_2^2. \tag{27}$$

In both cases, the additional term acts as a regularizer, simply using norms. However, when we directly add this term to attraction and repulsion shapes, we can easily explain what each term is doing. For the attractive term, it is increasing far-sightedness, whereas for the repulsive term, it adds a constant repulsive coefficient.

B More on UMAP's Learning Rate

As discussed in Sections 5 and 6 of the main text, UMAP is known to require learning rate annealing (Fig 6). To explain this, in Section 5 we defined the minimum distance for contraction (ζ_{-1}) and established that driving this value down through learning rate annealing produces compact clusters. In Section 6 we then compared the attraction shapes of UMAP (for $a = 1.0$ and $b = 1.0$) and Neg- t -SNE and showed that Neg- t -SNE tolerates a constant learning rate of 1.0 better than UMAP precisely because its attraction shape lies within $[-1, 0]$ while UMAP's spans $[-2, 0]$. By the same logic, we demonstrated that UMAP tolerates a constant learning rate of 0.5 once its attraction shape is confined to $[-1, 0]$ (Figs. 11 f,h).

Even so, the embeddings are consistently better when the learning rate anneals (across a wide range of a and b). The reason is direct: the algorithm's goal is to reduce ζ_{-1} to zero, which happens only as the learning rate approaches zero. Otherwise the embedding stays diffuse (Fig 6 a)). Conversely, if the learning rate starts too low, attraction and repulsion are both too weak and no clear clusters form (Fig 6 b)). The ζ_{-1} vs λ curve (Fig 6 c)) pinpoints a near-optimal operating point at $\lambda = 0.1$, where ζ_{-1} is low yet attraction and repulsion remain substantial. Fixing the learning rate at 0.1 indeed yields compact clusters with clean boundaries (Fig 6 d)).

C Large Language Model Embedding

Increased attraction and repulsion reliably produce better embeddings. To demonstrate this, we randomly selected 300,000 samples of PubMedBERT (Gu et al. 2021) features with known labels from the PubMed dataset (González-Márquez et al. 2024). Default UMAP yields a crowded structure in which labels (colors) overlap (Fig. 7 a)); stripped of labels, the embedding is baffling, since explicit clusters are absent. Adjusting b to 0.6 (as prescribed in Section 5.2) increases repulsion and, at the same time, promotes cluster formation through stronger attraction. The resulting mapping is readily explorable even without explicit labeling (Fig. 7 b)). This improvement is quantified by the k-NN accuracy ($k = 10$): on a hold-out set of 10,000 points, reducing b to 0.6 raises k-NN accuracy from 49.2% to 55.48%.

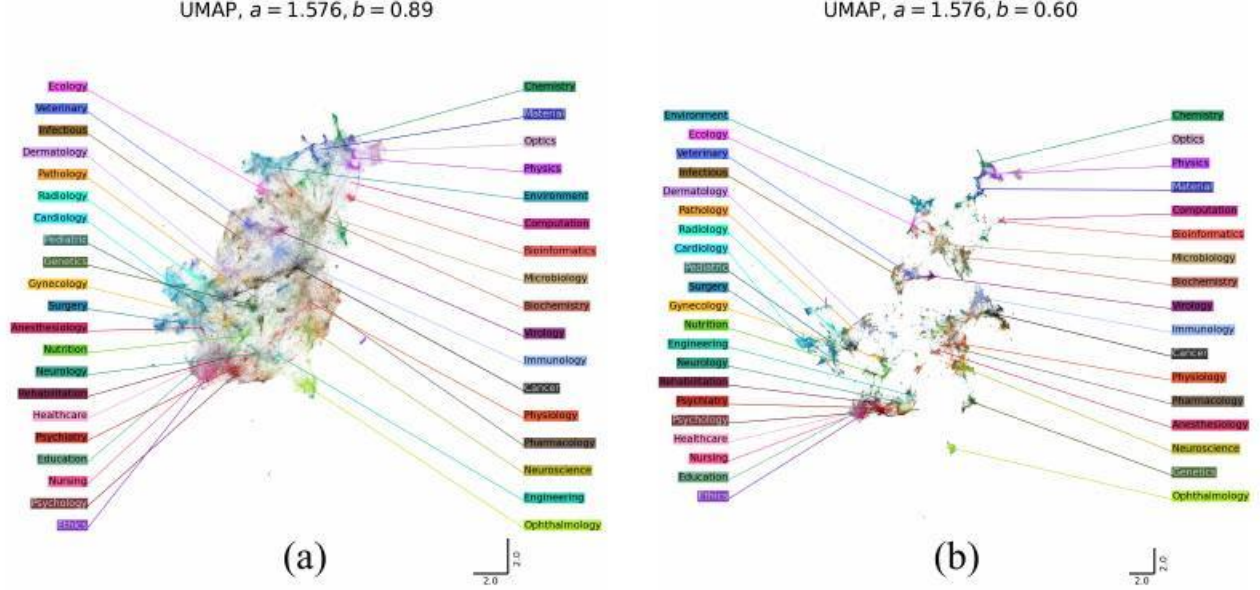


Figure 7: Visualization of 300K abstracts from the PubMed dataset. (a) Default UMAP ($a = 1.576, b = 0.89$) leaves labels overlapping throughout the mapping, whereas (b) UMAP with improved repulsion and attraction ($a = 1.576, b = 0.60$) produces distinctly better cluster formation and non-overlapping labels. Both mappings are labeled for clarity.

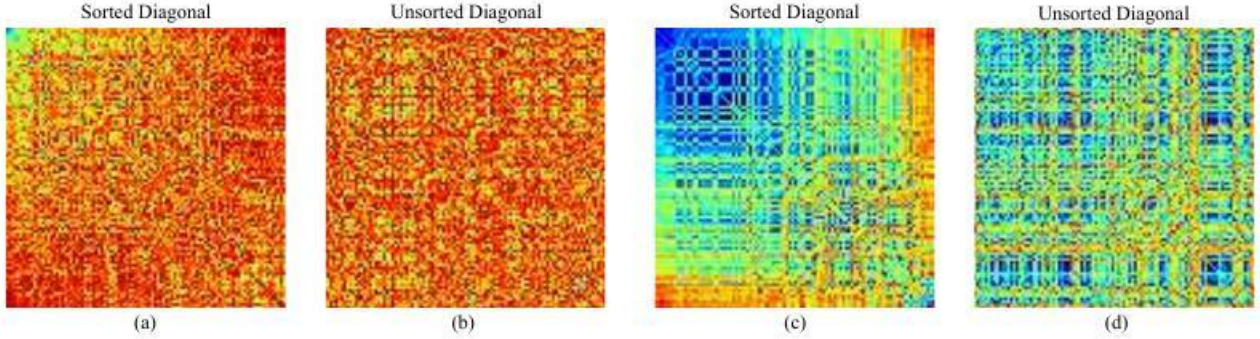


Figure 8: Effect of sorting the diagonal of Procrustes matrix on its visualization. (a) Procrustes matrix reproduced from [2]f) for the default UMAP attraction shape. The diagonal of the matrix is sorted by p_d of a sample embedding with PCA initialization. (b) The same data as in (a), but the diagonal is unsorted (or randomly sorted). (c) Procrustes matrix reproduced from [2]g) for modified attraction shape, where the diagonal of the matrix is sorted by p_d of a sample embedding with PCA initialization. (d) The same data as in (c), but the diagonal is unsorted.

D Metrics

D.1 Procrustes Distance and Procrustes Matrix

The Procrustes distance (Gower [1975]) measures the similarity between two point clouds $\{x\}$ and $\{y\}$ under linear transformations—translation, scaling, and rotation. Operationally, we hold the former fixed and vary the latter until the two sets are maximally aligned. Let $\{y'\}$ be the transformation of $\{y\}$ that achieves this. The Procrustes distance is then

$$p_d(\{x\}, \{y\}) = \sqrt{\sum_k (x_k - y'_k)^2}. \quad (28)$$

The Procrustes distance is a linear measure that has proven its worth across a wide range of settings (McInnes et al. [2018], Islam and Fleischer [2022], Kotlov et al. [2024]).

Here, we use the Procrustes distance to measure the consistency of embedding under random initialization. Let $\{x\}_p$ be a reference embedding (using PCA initialization in our experiments), and $X_r = \{\{x\}_i | i = 1, 2, 3, \dots, N\}$ be a set of

N embeddings obtained from random initialization. The similarity of the embeddings can be quantified by taking a mean and standard deviation of the strictly lower triangular values of the matrix P (reported as mean $\pm$ std in Figs. 2, 9 and 10), with

$$\text{mean} = \frac{2}{N(N-1)} \sum_{i,j(i>j)} P_{i,j}, \text{ and } \text{std} = \sqrt{\sum_{i,j(i>j)} \frac{2(P_{i,j} - \text{mean})^2}{N(N-1)}} \quad (29)$$

The indexes of X_r can be sorted such that $p_d(\{x\}_i, \{x\}_p) \leq p_d(\{x\}_{i+1}, \{x\}_p)$, so that the diagonal values of the Procrustes matrix are given by

$$P_{i,i} = p_d(\{x\}_i, \{x\}_p) \quad (30)$$

and the off-diagonal values are given by

$$P_{i,j} = p_d(\{x\}_i, \{x\}_j). \quad (31)$$

Numerically, sorting the diagonal adds little; visually, however, the ordering lays bare the underlying self-similarity of the embeddings. In Fig. 8(a), for instance, the sorted diagonal reveals that similar embeddings cluster in the upper-left region of the matrix. With a modified attraction shape, the number of mutually similar embeddings grows (the enlarged blue region in Fig. 8(c)), signaling a metastable point in the embedding algorithm. When the diagonal is left unsorted, this region vanishes and any sense of similar embeddings is lost (Fig. 8(b,d)).

D.2 Trustworthiness

The trustworthiness metric (Venna and Kaski 2001) quantifies how well local neighborhoods are preserved after dimensionality reduction:

$$T = 1 - \frac{2}{nk(2n-3k-1)} \sum_{i=1}^n \sum_{y_j \in \text{KNN}(y_i, k)} \max(0, r(i, j) - k) \quad (32)$$

where $\text{KNN}(y_i, k)$ is the k -NN graph in the embedding space and $r(i, j)$ is the rank of x_j in the high-dimensional k -NN graph. We set $k = 5$, assessing preservation of each point's five nearest neighbors. For computational efficiency, whenever we report trustworthiness we randomly sample 10,000 indices, and we use those same indices across every embedding we compare.

D.3 Silhouette Score

Although the silhouette score (Rousseeuw 1987) was designed to assess clustering algorithms, we use it to measure label separation in the embeddings—that is, how cleanly the ground-truth labels are clustered. The rationale is straightforward: embedding algorithms naturally produce clusters and should separate the labels as much as possible. For a point y_i in a point cloud $\{y\}$, let a_i be the mean distance from y_i to other points sharing its label and b_i the minimal mean distance from y_i to points of any other label. The pointwise silhouette is then

$$s_i = \frac{b_i - a_i}{\max\{a_i, b_i\}} \quad (33)$$

This value lies in $[-1, 1]$: close to 1 means y_i sits squarely within its own label cluster, near 0 marks a boundary point, and close to -1 signals failed label separation. The overall silhouette score is

$$\text{SIL} = \frac{1}{N} \sum s_i. \quad (34)$$

We computed the silhouette score for the whole embedding (no random sampling) using Euclidean distances.

E Effect of Random Initialization in Additional Datasets

The main text (Section 5.1) reported results only on MNIST. Here we run the identical experiment on Fashion-MNIST (FMNIST) (Xiao et al. 2017) and single-cell transcriptomes (Macosko et al. 2015) (Figs. 9 and 10 respectively). The conclusion holds without exception: modified and composite attraction shapes—those that increase attraction at large distances—significantly improve the consistency of reconstruction.

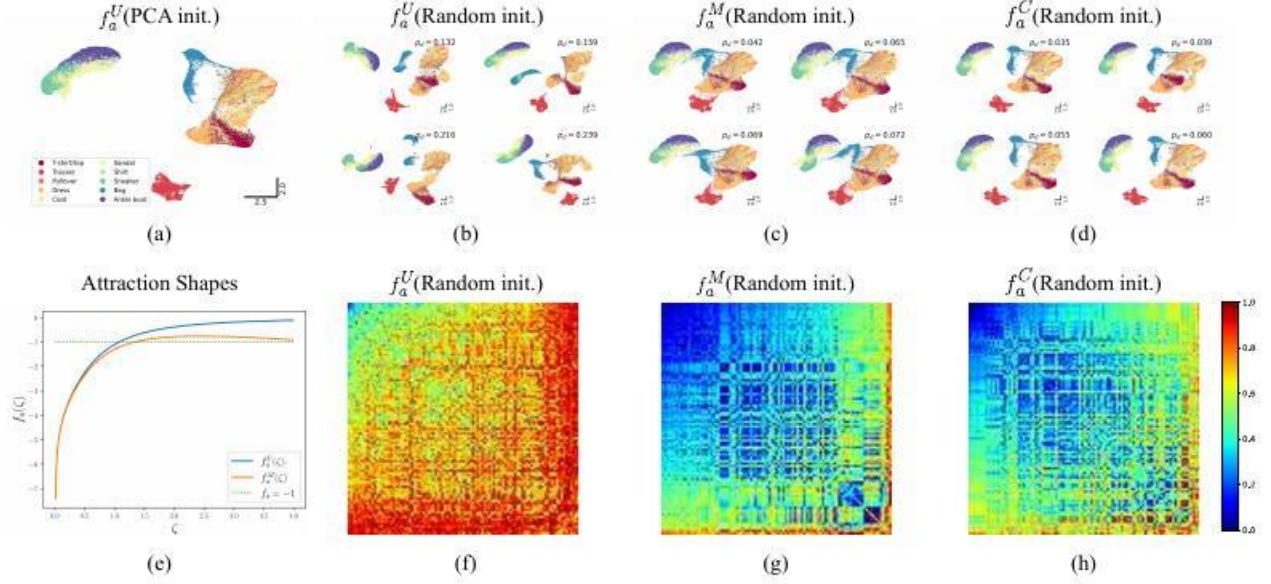


Figure 9: Effect of random UMAP initialization across attraction shapes on FMNIST. (a) PCA mapping, used as the standard. (b-d) The four mappings with the lowest Procrustes distance (p_d) from the embedding in (a) for (b) default, (c) modified, and (d) composite attraction shapes. (e) Default UMAP and modified attraction shapes. (f-h) Procrustes matrix from 100 runs of (f) default (0.72 ± 0.15), (g) modified (0.38 ± 0.19), and (h) composite (0.43 ± 0.18) attraction shapes.

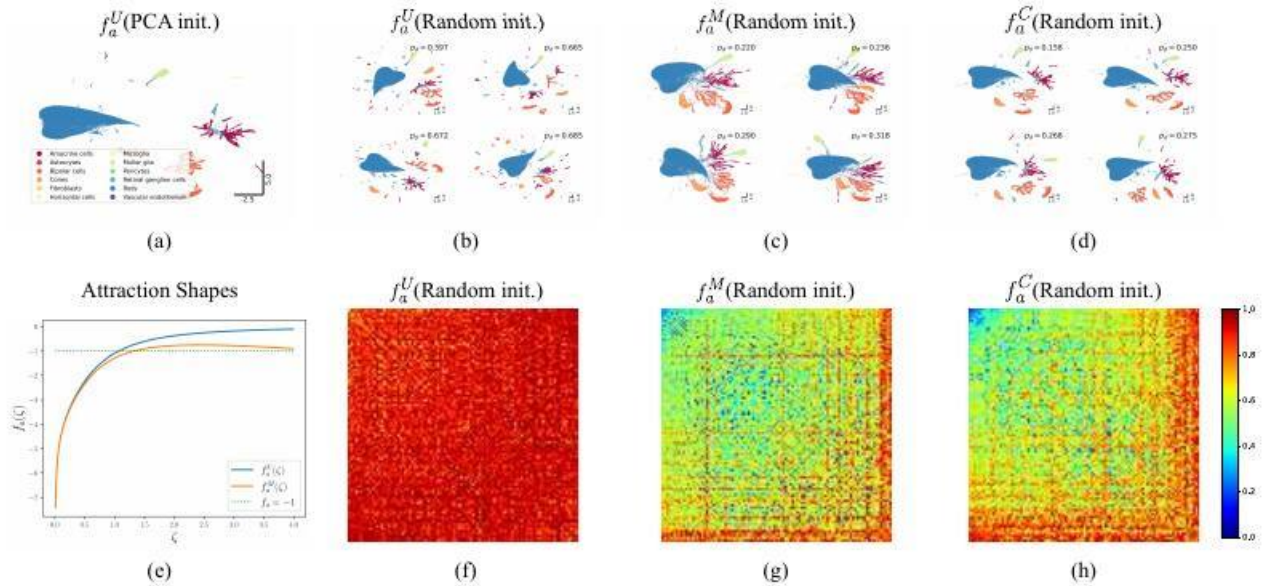


Figure 10: Effect of random UMAP initialization across attraction shapes on single-cell transcriptomes data. (a) PCA mapping, used as the standard. (b-d) The four mappings with the lowest Procrustes distance (p_d) from the embedding in (a) for (b) default, (c) modified, and (d) composite attraction shapes. (e) Default UMAP and modified attraction shapes. (f-h) Procrustes matrix from 100 runs of (f) default (0.91 ± 0.06), (g) modified (0.61 ± 0.13), and (h) composite (0.64 ± 0.15) attraction shapes.

F More on NEG-*t*-SNE

F.1 Comparison to UMAP

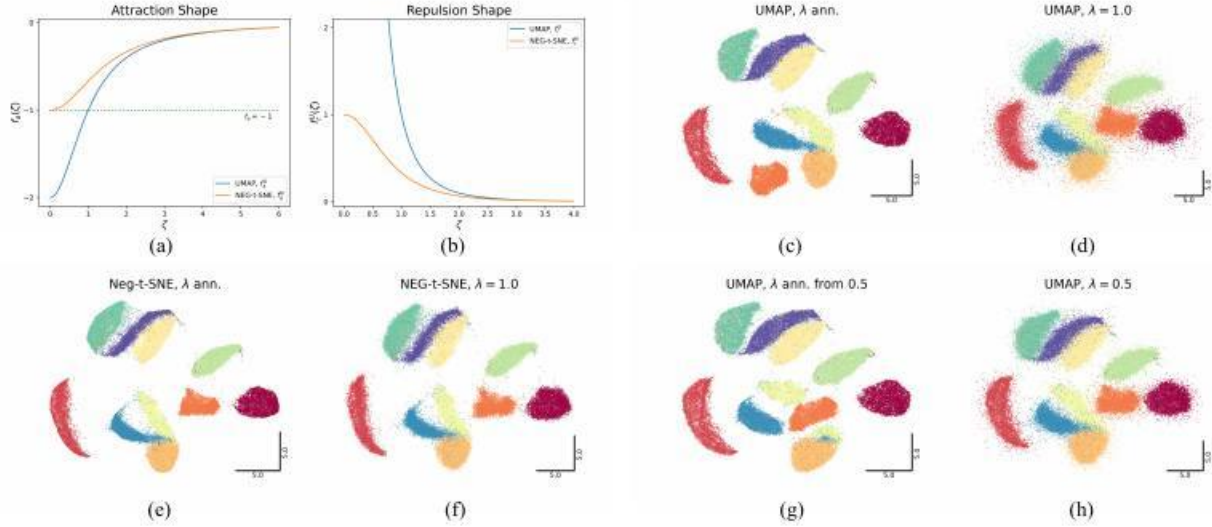


Figure 11: Sensitivity of UMAP and NEG-*t*-SNE to the learning rate on the MNIST dataset. (a) Attraction and (b) repulsion shapes for UMAP ($a = 1, b = 1$) and NEG-*t*-SNE. (c,d) UMAP is highly sensitive to the learning rate λ , since $f_a^U < -1$ as the separation distance ζ decreases; without annealing, the clusters turn fuzzy. (e,f) NEG-*t*-SNE is far less sensitive to λ because $f_a^N \in [-1, 0]$ always, so its clusters stay crisp even without annealing. (g,h) Confining f_a^U to $[-1, 0]$ by setting $\lambda = 0.5$ likewise reduces sensitivity to λ .

Figure 11 shows the shapes of UMAP and NEG-*t*-SNE alongside various MNIST embeddings. Under annealing, UMAP and NEG-*t*-SNE produce similar output (Figs. 11(c,e)). At a constant learning rate of 1, however, UMAP turns fuzzy while NEG-*t*-SNE retains much cleaner boundaries (Fig. 11(d,f)). The argument in Section 6 predicts that confining f_a^U to $[-1, 0]$ yields less fuzzy clusters at fixed λ . Fig. 11(h,i) already showed this, with UMAP delivering better embedding and clustering when the attraction learning rate (λ_a) was $\lesssim 0.5$. The simplest way to enforce it is to set $\lambda = 0.5$, which satisfies Proposition 4.1 for all ζ . The resulting embeddings in Figs. 11(g) and (h) confirm the prediction: the clusters match those of NEG-*t*-SNE, and at constant $\lambda = 0.5$ they are far less fuzzy than before, with sharper boundaries (Fig. 11(d,h)). A few points still stray outside the clusters because of UMAP’s repulsion shape—exactly the issue NEG-*t*-SNE resolves.

Next, we can introduce the parameters a and b into NEG-*t*-SNE (essentially the formulation of Parametric UMAP; see Proposition 6.1). The affinity function becomes $q_{ij}^N = 1/(2 + a\zeta^{2b})$, and the attraction and repulsion shapes become $f_a^N = -\frac{2ab\zeta^{2(b-1)}}{2+a\zeta^{2b}}$, and $f_r^N = \frac{2ab\zeta^{2(b-1)}}{(1+a\zeta^{2b})(2+a\zeta^{2b})}$, respectively. For $0 < b < 1$, both shapes become unbounded as $\zeta \rightarrow 0$. NEG-*t*-SNE therefore inherits exactly the numerical challenges of UMAP once a and b vary, along with the corresponding limitations. One notable distinction is that, relative to UMAP, its attraction shape attains a lower minimum distance for contraction (ζ_{-1}). This may aid cluster formation, but it also decays to zero faster (heightened near-sightedness with distance), which can blunt attraction over longer distances.

F.2 Comparison to Parametric UMAP

Parametric UMAP was originally trained with the UMAP objective (Sainburg et al., 2021), but later work adopted a numerically stable, log-sigmoid-based modified cross-entropy loss (Shi et al., 2023). This modification makes Parametric UMAP and Neg-t-SNE (Damrich et al., 2023) exactly equivalent (Proposition 6.1). We prove the equivalence below.

Proof of Proposition 6.1

Proof. The kernel function under this modification becomes

$$q_{ij}^P = -\log(1 + a\|y_i - y_j\|_2^{2b}). \quad (35)$$

The attractive term is

$$-\text{logsigmoid}(q_{ij}^P) = -\log\left(\frac{1}{1 + \exp(-q_{ij}^P)}\right) \quad (36)$$

$$= -\log\left(\frac{1}{2 + a\|y_i - y_j\|_2^{2b}}\right) \quad (37)$$

$$= -\log(q_{ij}^N). \quad (38)$$

And the repulsive term is

$$\text{logsidmoid}(q_{ij}^P) - q_{ij}^P = \log\left(\frac{1}{2 + a\|y_i - y_j\|_2^{2b}}\right) + \log(1 + a\|y_i - y_j\|_2^{2b}) \quad (39)$$

$$= \log\left(\frac{1 + a\|y_i - y_j\|_2^{2b}}{2 + a\|y_i - y_j\|_2^{2b}}\right) \quad (40)$$

$$= \log\left(1 - \frac{1}{2 + a\|y_i - y_j\|_2^{2b}}\right) \quad (41)$$

$$= \log(1 - q_{ij}^N). \quad (42)$$

Both these are NEG-t-SNE with explicit parameters a and b (while in Neg-t-SNE these are set to 1). $\square$

G Alternate Dimensionality Reduction Algorithms

The alternative algorithms we consider use the same kernel function as UMAP (with $a = 1$ and $b = 1$ in their low-dimensional weight):

$$q_{ij} = \frac{1}{1 + \|y_i - y_j\|_2^2}. \quad (43)$$

We begin with the TriMap (Amid and Warmuth [2019]) algorithm. Although TriMap relies on triplets rather than pairwise interactions, it serves as an ideal primer for analyzing attraction and repulsion that are more involved than UMAP's. We then treat Pairwise Controlled Manifold Approximation (PaCMAP) (Wang et al. [2021]) and its extension, Pairwise Controlled Manifold Approximation with Local Adjusted Graph (LocalMAP) (Wang et al. [2025]), both of which recast TriMap's loss for pairwise interactions. Next we tackle t -SNE (Van der Maaten and Hinton [2008]), add a short note on SNE (Hinton and Roweis [2002]) with its alternate kernel, and close with a brief treatment of multidimensional scaling (Borg and Groenen [2007]).

G.1 TriMap

TriMap (Amid and Warmuth [2019]) optimizes low-dimensional using a triplet loss

$$\mathcal{L}^T = \sum_{(i,j,k)} w_{ijk} \frac{1}{1 + \frac{q_{ij}}{q_{ik}}}, \quad (44)$$

where w_{ijk} is the weight of the triplet (y_i, y_j, y_k) , y_j lies in the high-dimensional k -nearest neighbor set of y_i , and y_k is a far-away point. Minimizing this loss drives y_i and y_j to attract while y_i and y_k repel. The update equations are

$$y_i^{t+1} = y_i^t + \lambda f_a^T(\zeta_1^t, \zeta_2^t)(y_i^t - y_j^t) + \lambda f_r^T(\zeta_1, \zeta_2)(y_i^t - y_k^t), \quad (45)$$

$$y_j^{t+1} = y_j^t - \lambda f_a^T(\zeta_1^t, \zeta_2^t)(y_i^t - y_j^t), \quad (46)$$

$$y_k^{t+1} = y_k^t - \lambda f_r^T(\zeta_1^t, \zeta_2^t)(y_i^t - y_k^t), \quad (47)$$

where $\zeta_1 = \|y_i - y_j\|_2$ is the nearest-neighbor distance, $\zeta_2 = \|y_i - y_k\|_2$ is the distance to the faraway point, and f_a^T and f_r^T are TriMap's attraction and repulsion shapes, respectively. Unlike UMAP, the attractive and repulsive components are non-separable, and the shapes depend on two distance measures (making them 2D). The attraction shape is

$$f_a^T(\zeta_1, \zeta_2) = -\frac{2(1 + \zeta_2^2)}{(2 + \zeta_1^2 + \zeta_2^2)^2}, \quad (48)$$

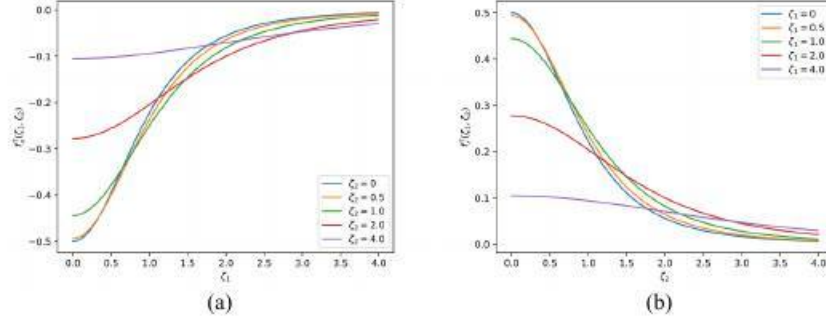


Figure 12: (a) TriMap attraction shapes for different ζ_2 and (b) TriMap repulsion shapes for different ζ_1 .

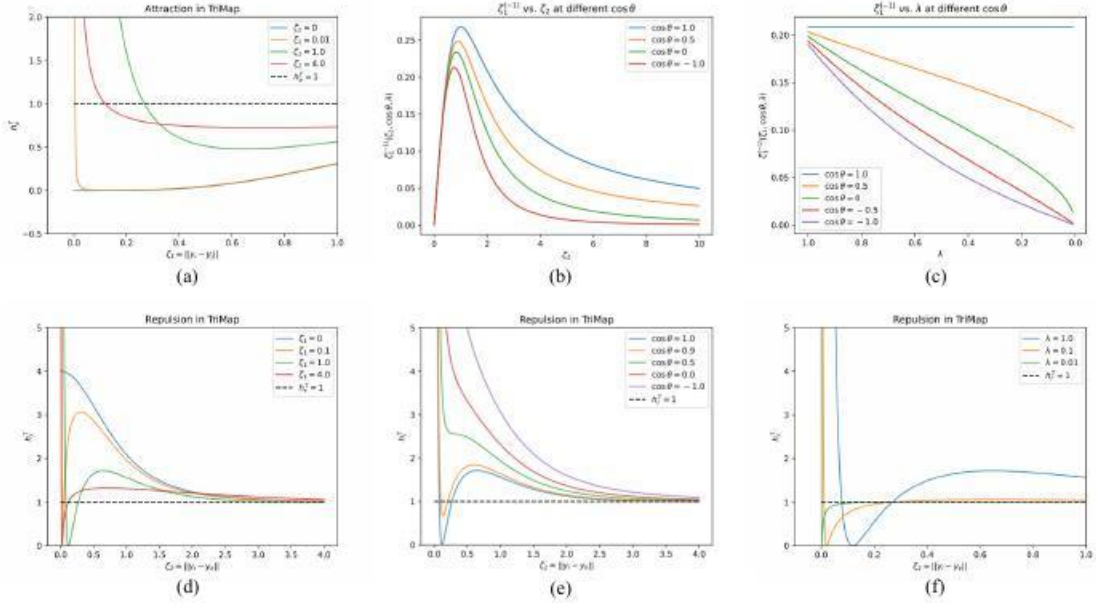


Figure 13: Attraction and repulsion behavior in TriMap. (a) h_a^T vs ζ_1 for different ζ_2 . Values below the dotted line indicate attraction. Like UMAP, TriMap exhibits both attraction and repulsion for nearest neighbors (y_i, y_j). (b,c) Unlike UMAP, the minimum distance for contraction ($\zeta_1^{(-1)}$) shifts with (b) ζ_2 and (c) λ ; $\cos \theta$ sets the range of these values. (d-f) Repulsion in TriMap as we vary (d) ζ_1 , (e) $\cos \theta$, and (f) λ . Values above (below) the dotted line indicate repulsion (attraction). Whereas UMAP's repulsive force only ever repels, TriMap's can both attract and repel. Unless otherwise labeled, $\zeta_1 = 1.0$, $\zeta_2 = 0.5$, $\cos \theta = 1.0$, and $\lambda = 1.0$.

and the repulsion shape is

$$f_r^T(\zeta_1, \zeta_2) = \frac{2(1 + \zeta_1^2)}{(2 + \zeta_1^2 + \zeta_2^2)^2}. \quad (49)$$

TriMap’s attraction and repulsion shapes (Fig. 12) follow trends similar to UMAP’s. Its repulsion shape bottoms out at -0.5 , so—unlike UMAP—attraction alone never flips positions. Because Eqs. (45)–(47) do not decouple the attractive and repulsive terms, Propositions 4.1 and 4.2 no longer apply directly. Focusing on attraction first, we prove

Proposition G.1. *Update equations (45)–(47) provide a contraction if*

$$h_a^T(\zeta_1, \zeta_2, \theta, \lambda) < 1, \quad (50)$$

where $h_a^T(\zeta_1, \zeta_2, \theta, \lambda) = (1 + 2\lambda f_a^T)^2 + 2(1 + 2\lambda f_a^T)\lambda f_r^T \frac{\zeta_2}{\zeta_1} \cos \theta + (\lambda f_r^T)^2 \frac{\zeta_2^2}{\zeta_1^2}$ and θ is the angle between the vectors $(y_i - y_j)$ and $(y_i - y_k)$, i.e., $\cos \theta = \frac{(y_i^t - y_j^t)^T (y_i^t - y_k^t)}{\|y_i^t - y_j^t\|_2 \|y_i^t - y_k^t\|_2}$.

Proof. We require

$$\|y_i^{t+1} - y_j^{t+1}\|_2^2 < \|y_i^t - y_j^t\|_2^2. \quad (51)$$

From Eq. (45) and (46): $y_i^{t+1} - y_j^{t+1} = (1 + 2\lambda f_a^T)(y_i^t - y_j^t) + \lambda f_r^T (y_i^t - y_k^t)$. Putting this value in Eq. (51), we obtain the desired inequality. $\square$

Inequality (50) depends on ζ_1 , ζ_2 , $\cos \theta$, and λ . Crucially, the value of $\zeta_1 = \|y_i - y_j\|_2$ where we seek contraction is now coupled to additional variables. Figure 13(a) shows the attraction behavior for various ζ_2 at $\cos \theta = 1$ and $\lambda = 1$. Values below the dotted line indicate attraction (contraction of ζ_1); values above indicate repulsion (expansion of ζ_1). Where the dotted line meets h_a^T we obtain the minimum distance for contraction ($\zeta_1^{(-1)}$), the direct analog of UMAP’s ζ_{-1} , defined as

$$\zeta_1^{(-1)}(\zeta_2, \theta, \lambda) = \arg \min_{\zeta_1} |h_a^T(\zeta_1, \zeta_2, \theta, \lambda) - 1|, \quad (52)$$

$$\text{s.t. } \zeta_1 \geq 0. \quad (53)$$

$\zeta_1^{(-1)}$ is finite and strictly positive in most cases (Fig. 13(b,c)). TriMap thus behaves much like UMAP and requires learning rate annealing or an equivalent mechanism (in practice, the TriMap authors use the delta-bar-delta (Jacobs, 1988) method under appropriate initialization).

Proposition G.2. *Update equations (45)–(47) provide expansion if*

$$h_r^T(\zeta_1, \zeta_2, \theta, \lambda) > 1, \quad (54)$$

where $h_r^T(\zeta_1, \zeta_2, \theta, \lambda) = (1 + 2\lambda f_r^T)^2 + 2\lambda f_a^T (1 + \lambda f_r^T) \frac{\zeta_1}{\zeta_2} \cos \theta + (\lambda f_a^T)^2 \frac{\zeta_1^2}{\zeta_2^2}$.

Proof. We require

$$\|y_i^{t+1} - y_k^{t+1}\|_2^2 > \|y_i^t - y_k^t\|_2^2. \quad (55)$$

From Eq. (45) and (47): $y_i^{t+1} - y_k^{t+1} = (1 + 2\lambda f_r^T)(y_i^t - y_k^t) + \lambda f_a^T (y_i^t - y_j^t)$. Putting this value in Eq. (55) we obtain inequality (54). $\square$

Inequality (54) likewise depends on ζ_1 , ζ_2 , $\cos \theta$, and λ . Here we care about the expansion of $\zeta_2 = \|y_i - y_k\|_2$. Figures 13(d-f) show the repulsion behavior as the other quantities vary. Values above the dotted line indicate repulsion (expansion of distance); values below indicate attraction. The striking contrast with UMAP is that TriMap’s repulsion can produce contraction rather than expansion. Since this anomaly is confined to small distances, appropriate initialization and triplet selection eliminate it.

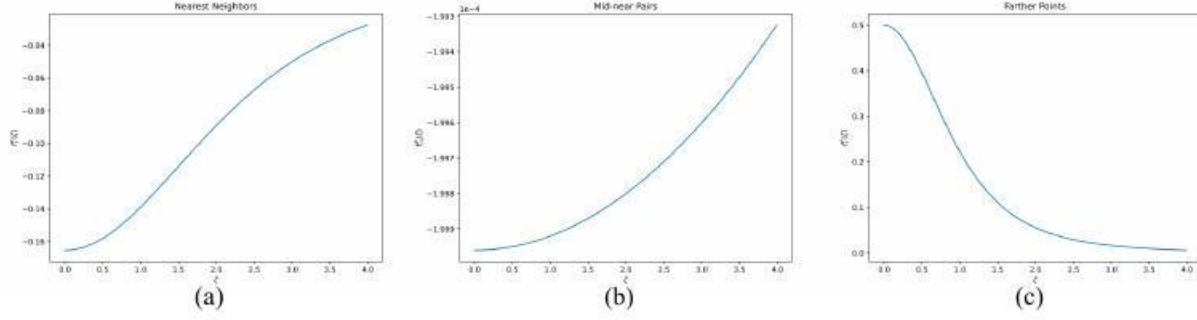


Figure 14: Attraction and repulsion shapes of PaCMAP. (a,b) Attraction shapes for (a) nearest-neighbor and (b) mid-near points (note that the values are on the order of 10^{-4}). (c) Repulsion shape for farther pairs. $\lambda = 1$ for all figures.

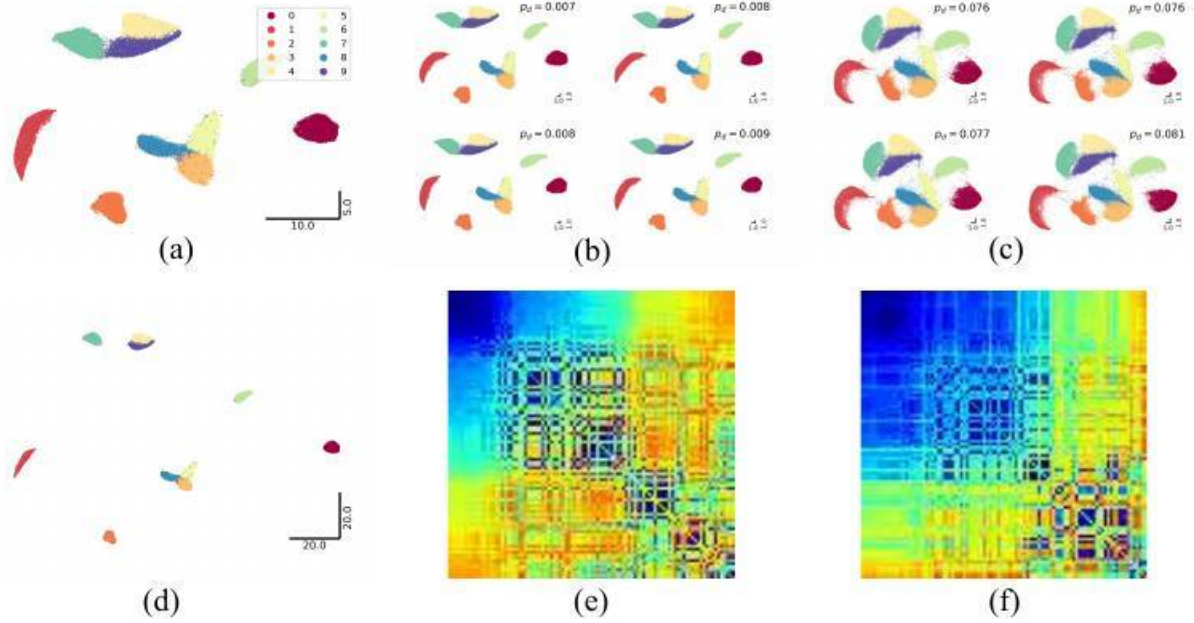


Figure 15: PaCMAP behavior under different conditions. (a) PaCMAP of MNIST with PCA initialization. (b,c) The four samples best matching (a) for (b) original PaCMAP and (c) PaCMAP with the modified attraction shape $f_a^M(\zeta) = f_a^P(\zeta) - 0.001\zeta$, under random initialization. (d) PaCMAP of MNIST with the modified repulsion shape $f_r^M = f_r^P + 0.00005$. (e,f) Procrustes matrix for (e) original PaCMAP (0.51 ± 0.22) and (f) PaCMAP with the modified attraction shape (0.43 ± 0.22). Exactly as in UMAP, stronger attraction at farther distances improves consistency, while stronger repulsion yields smaller clusters and larger inter-cluster distances.

G.2 Pairwise Controlled Manifold Approximation (PaCMAP)

PaCMAP (Wang et al. 2021) optimizes low-dimensional embedding at different scales. The loss function is

$$\mathcal{L}^P = w_{NB} \sum_{(i,j) \in NN} \frac{1}{1 + 10q_{ij}} + w_{MN} \sum_{(i,k) \in MN} \frac{1}{1 + 10000q_{ik}} + w_{FP} \sum_{(i,l) \in FP} \frac{1}{1 + \frac{1}{q_{il}}}, \quad (56)$$

where w_{NB} , w_{MN} , and w_{FP} are the weights of the nearest neighbor (NN) pairs, mid-near (MN) pairs, and further pairs (FP), respectively (details in Appendix H). The first two terms attract, the last repels. The loss is, in fact, a modified TriMap triplet loss: for the attractive terms it swaps TriMap’s affinity for distant points with constants (1/10 for nearest neighbors, 1/10000 for mid-near pairs, and 1 for farther points). The loss thus splits into three separable terms and

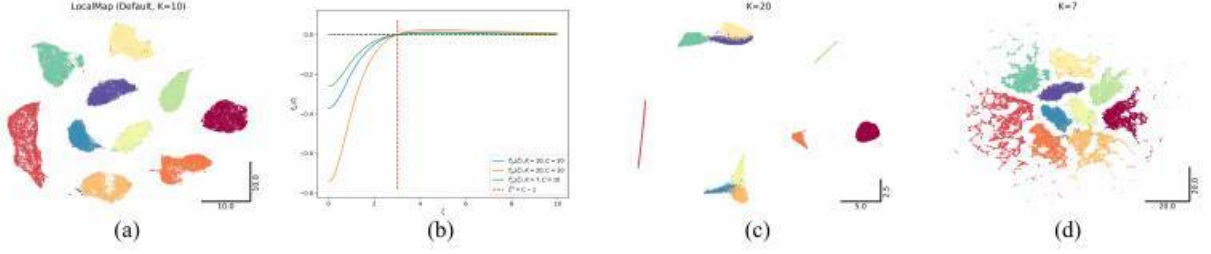


Figure 16: Behavior of LocalMAP on MNIST. (a) Default embedding. (b) Attraction-repulsion shape of nearest neighbors. Values below (above) the dotted line indicate attraction (repulsion); the transition from attraction to repulsion occurs as ζ grows past $\sqrt{C-1}$. Following the LocalMAP implementation, we set $C = 10$. (c) At large $K (= 20)$, repulsion dominates and clusters grow compact. (d) At small $K (= 7)$, attraction dominates and clusters break apart.

decouples the update equations. The nearest-neighbor updates are

$$y_i^{t+1} = y_i^t + \lambda f_a^P(\zeta_1^t)(y_i^t - y_j^t), \quad (57)$$

$$y_j^{t+1} = y_j^t - \lambda f_a^P(\zeta_1^t)(y_i^t - y_j^t), \quad (58)$$

of the mid-near pairs are

$$y_i^{t+1} = y_i^t + \lambda f_m^P(\zeta_2^t)(y_i^t - y_k^t), \quad (59)$$

$$y_k^{t+1} = y_k^t - \lambda f_m^P(\zeta_2^t)(y_i^t - y_k^t), \quad (60)$$

and of the farthest pairs are

$$y_i^{t+1} = y_i^t + \lambda f_r^P(\zeta_3^t)(y_i^t - y_l^t), \quad (61)$$

$$y_l^{t+1} = y_l^t - \lambda f_r^P(\zeta_3^t)(y_i^t - y_l^t), \quad (62)$$

where $\zeta_1 = \|y_i - y_j\|_2$, $\zeta_2 = \|y_i - y_k\|_2$, $\zeta_3 = \|y_i - y_l\|_2$ are distances, f_a^P and f_m^P are attraction shapes for nearest neighbors and mid-near pairs, respectively, and f_r^P is the repulsion shape for the farthest pairs. Correspondingly, the functional forms of the shapes are

$$f_a^P(\zeta) = -\frac{20}{(11 + \zeta^2)^2}, \quad (63)$$

$$f_m^P(\zeta) = -\frac{20000}{(10001 + \zeta^2)^2}, \quad (64)$$

$$f_r^P(\zeta) = \frac{2}{(2 + \zeta^2)^2}. \quad (65)$$

f_a^P and f_m^P obey Proposition 4.1 and f_r obeys Proposition 4.2 (Fig. 14). The attraction is much weaker than UMAP’s, yet ample across a wide range of learning rates (modulated by Adam (Kingma and Ba 2015)); with $w_{NB} = 3$ the maximum attraction stays below 0.5, ruling out any flips during the attractive update. PaCMAP typically initializes the embedding within a small low-dimensional sphere (e.g., on the order of 10^{-3} in (Wang et al. 2021)) and relies on repulsion to pull the clusters apart. It largely recovers UMAP’s clustering properties (especially on MNIST) with improved ordering, and its consistency under random initialization already exceeds UMAP’s (Fig. 15(b,e))—consistency we improve still further with a modified attraction (Fig. 15(c,f)). Adding a small value to the repulsion shape strengthens repulsion and makes the embedding more compact by enlarging inter-cluster distance.

G.3 Pairwise Controlled Manifold Approximation with Local Adjusted Graph (LocalMAP)

LocalMAP (Wang et al. 2025) is a successor to PaCMAP. Its defining feature is the clean separation of all 10 MNIST clusters (Fig. 16(a)), mirroring the behavior in Figs. 3(g,h,j,k) and Figs. 4(a,b). Here we dissect how attractive and repulsive forces jointly control the compactness and connectedness of clusters.

LocalMAP runs PaCMAP and then performs additional optimization on the attraction to decouple certain clusters, minimizing the loss function

$$\mathcal{L}^L = \sum_{(i,j) \in NN} \frac{K}{\frac{1}{\sqrt{q_{ij}}} + C\sqrt{q_{ij}}} + \sum_{(i,l) \in FP} \frac{1}{1 + \frac{1}{q_{il}}}. \quad (66)$$

The first term fuses the attractive and repulsive character of the triplet loss into a single expression acting on the same pair: in one regime it attracts, in the other it repels. The second term is identical to PaCMAP’s, save that the algorithm resamples further pairs every few iterations. We therefore analyze only the first, nearest-neighbor term. The update equations are

$$y_i^{t+1} = y_i^t + \lambda f_{ar}^L(\zeta_i^t)(y_i^t - y_j^t), \quad (67)$$

$$y_j^{t+1} = y_j^t - \lambda f_{ar}^L(\zeta_j^t)(y_i^t - y_j^t), \quad (68)$$

where f_{ar}^L is the attraction-repulsion shape given by

$$f_{ar}^L(\zeta) = -\frac{K(C-1-\zeta^2)}{2\sqrt{1+\zeta^2}(1+C+\zeta^2)^2}. \quad (69)$$

The update contracts as long as $\zeta^2 < C-1$ and $-1 < \lambda f_{ar}^L < 0$ (from Proposition 4.1). Once $\zeta^2 > C-1$, $f_{ar} > 0$, and by Proposition 4.2 the update expands. K and C together decide whether attraction or repulsion dominates. The LocalMAP implementation sets both C and K to 10 (Fig. 16(b)), giving stronger attraction than PaCMAP (Fig. 14(a)). Consequently, once $\zeta > 3$ the nearest neighbors repel, and pairs bridging two clusters separate. K , acting as a scaling parameter for the forces, controls this separation.

By Proposition 4.1 the condition $\lambda f_{ar}^L(0) \geq -\frac{1}{2}$ pins down the maximum K —and the maximum attraction f_{ar}^L can supply—without flipping the pairs. (We want to avoid such flips during LocalMAP optimization to preserve the ordering from PaCMAP, since flipping would inhibit cluster separation.) This reduces (69) to $K \leq \frac{\lambda(1+C)^2}{C-1}$, which for $C = 10$ and $\lambda = 1$ gives $K \lesssim 13.44$. For $K \gtrsim 13.44$, repulsive forces dominate and clusters grow more compact, but LocalMAP’s objective fails: the bridge between clusters persists (Fig. 16(c) for $K = 20$). As K decreases, attractive forces dominate (repulsion becomes too weak) and existing clusters break apart (Fig. 16(d) for $K = 7$, mirroring Fig. 4(b)).

G.4 t -distributed Stochastic Neighbor Embedding (t -SNE)

t -SNE (Van der Maaten and Hinton [2008]) optimizes pairwise distances. The loss function is

$$\mathcal{L} = -\sum_{i,j} w_{i,j} \log \left(\frac{q_{i,j}}{\sum_{k \neq l} q_{k,l}} \right), \quad (70)$$

where $w_{i,j}$ is the weight of the pair. The original implementation of t -SNE considers all the pairs (not just nearest neighbors). This loss function decomposes into attraction and repulsion forces by

$$\mathcal{L} = \sum_{i,j} \left[-w_{i,j} \log(q_{i,j}) + w_{i,j} \log \left(\sum_{k \neq l} q_{k,l} \right) \right]. \quad (71)$$

As before, the first term supplies the attractive forces and the second the repulsive forces. The attractive term is identical to UMAP’s (with $a = 1$ and $b = 1$) and cheap to compute, whereas the repulsive term couples every pair and is therefore expensive. Applying the same principles as for UMAP, we write t -SNE’s update equations. Because the original t -SNE dispenses with the nearest-neighbor graph, the weight $w_{i,j}$, computed over all pairs, plays a central role in the updates. The attractive update equations are

$$y_i^{t+1} = y_i^t + \lambda w_{i,j} f_a^{t-SNE}(\zeta_{i,j}^t)(y_i^t - y_j^t), \quad (72)$$

$$y_j^{t+1} = y_j^t - \lambda w_{i,j} f_a^{t-SNE}(\zeta_{i,j}^t)(y_i^t - y_j^t), \quad (73)$$

where f_a^{t-SNE} is the attraction shape of t -SNE and $\zeta_{i,j} = \|y_i - y_j\|_2$. The update equation for the repulsive parts are

$$y_i^{t+1} = y_i^t + \lambda \frac{w_{i,j}}{Z} \sum_k f_r^{t-SNE}(\zeta_{i,k}^t)(y_i^t - y_k^t), \quad (74)$$

$$y_j^{t+1} = y_j^t - \lambda \frac{w_{i,j}}{Z} \sum_l f_r^{t-SNE}(\zeta_{l,j}^t)(y_l^t - y_j^t), \forall j, j \neq i, \quad (75)$$

where f_r^{t-SNE} is the repulsion shape of t -SNE and $Z = \sum_{k \neq l} q_{k,l}$. The functional forms of these shapes are

$$f_a^{t-SNE}(\zeta) = -\frac{2}{1+\zeta^2}, \text{ and} \quad (76)$$

$$f_r^{t-SNE}(\zeta) = \frac{2}{(1+\zeta^2)^2}. \quad (77)$$

From the attractive update Eqs. (72)-(73), f_a^{t-SNE} obeys Proposition 4.1 (with $0 < \lambda w_{i,j} f_a^{t-SNE} < -1$) and yields the minimum distance for contraction, ζ_{-1} . The repulsive update couples all pairs and so bears no simple relation to the repulsion shape. Drawing on our TriMap analysis, however, we can derive the following:

Proposition G.3. *The update Eqs. (74)-(75) provide an expansion if*

$$h_r^{t-SNE}(\zeta_{i,j}, v, \theta, \lambda, w_{i,j}) > 1, \quad (78)$$

where

$$h_r^{t-SNE}(\zeta_{i,j}, v, \theta, \lambda, w_{i,j}) = (1 + 2\lambda \frac{w_{i,j}}{Z} f_r^{t-SNE}(\zeta_{i,j}))^2 + \frac{\|v\|_2^2}{\zeta_{i,j}^2} + 2\lambda \frac{w_{i,j}}{Z} f_r^{t-SNE}(\zeta_{i,j}) \frac{\|v\|_2}{\zeta_{i,j}} \cos \theta,$$

$$v = \lambda \frac{w_{i,j}}{Z} \left(\sum_{k,k \neq j} f_r^{t-SNE}(\zeta_{i,k})(y_i - y_k) + \sum_{l,l \neq i} f_r^{t-SNE}(\zeta_{l,j})(y_l - y_j) \right),$$

and θ is the angle between $(y_i - y_j)$ and v .

Proof. We require,

$$\|y_i^{t+1} - y_j^{t+1}\|_2^2 > \|y_i^t - y_j^t\|_2^2. \quad (79)$$

From Eqs. (74) and (75), $y_i - y_j = (1 + 2\lambda \frac{w_{i,j}}{Z} f_r^{t-SNE}(\zeta_{i,j}))(y_i - y_j) + v$. Putting this value in Eq. (79), we obtain the desired inequality. $\square$

Because t -SNE's repulsion condition (Eq. 78) mirrors TriMap's, the repulsion behavior is identical: t -SNE's repulsive forces can produce both attraction and repulsion.

Since t -SNE's forces are scaled (by $w_{i,j}$ for attraction and $\frac{w_{i,j}}{Z}$ for repulsion), both are typically weaker than UMAP's. The algorithm therefore relies on large learning rates (e.g., $\approx 10^3$ in the Open- t -SNE package (Poličar et al., 2024)). Moreover, most t -SNE implementations require an 'early exaggeration' step that multiplies the attractive forces by a constant for the first few iterations. This pulls together points that should be neighbors but currently sit far apart (inducing far-sightedness), and it simultaneously separates points that are too close ($\zeta_{i,j} \approx \zeta_{-1}$) yet ought to be apart. 'Early exaggeration' is thus indispensable for finding a consistent t -SNE embedding, above all under random initialization. (Applied throughout the optimization, this same trick drives t -SNE embeddings toward UMAP-like embeddings (Böhm et al., 2022).)

G.5 Stochastic Neighbor Embedding (SNE)

SNE (Hinton and Roweis, 2002) is one of the oldest algorithms in this class. It follows the same formula as t -SNE (Eq. 71) but with $q_{i,j} = \exp(-\|y_i - y_j\|_2^2)$, giving the attraction shape

$$f_a^{SNE}(\zeta) = -2, \quad (80)$$

that follows Proposition 4.1 (with $0 < \lambda w_{i,j} f_a^{SNE} < -1$) and the repulsion shape

$$f_r^{SNE}(\zeta) = 2 \exp(-\zeta^2), \quad (81)$$

which follows Proposition G.3 (replacing the t -SNE symbols with their SNE counterparts). The attraction shape is ill-posed and hence depends almost entirely on the learning rate (λ) and weights ($w_{i,j}$) for contraction, so clusters overlap even for a small number of samples—the crowding problem (Van der Maaten and Hinton, 2008). t -SNE and its successors fix this by abandoning the Gaussian kernel for a heavy-tailed one, Eq. (43).

G.6 Multidimensional Scaling (MDS)

Multidimensional scaling (MDS) (Borg and Groenen, 2007) usually avoids gradient methods, which often fail to converge to good mappings, and instead uses stress majorization. A handful of works nonetheless pursue gradient methods (Kruskal, 1964; Zheng et al., 2018), and we treat this case briefly below. In particular, Zheng et al. (2018) formulates a successful gradient-descent MDS algorithm for graph drawing. We start from the cost function (dropping the weight $w_{i,j}$ without loss of generality and folding it into the learning rate for the discussion):

$$\mathcal{L}^{MDS} = \sum_{i,j,i < j} (\|y_i - y_j\| - d_{ij})^2. \quad (82)$$

The loss has a single term per pair with no explicit attractive and repulsive terms; that one term supplies both attraction and repulsion. Fortunately, the loss separates into individual (i, j) components, so we can analyze a single pair. The derivative is

$$\frac{\partial}{\partial y_i} (\|y_i - y_j\| - d_{ij})^2 = 2 \frac{\|y_i - y_j\| - d_{ij}}{\|y_i - y_j\|} (y_i - y_j). \quad (83)$$

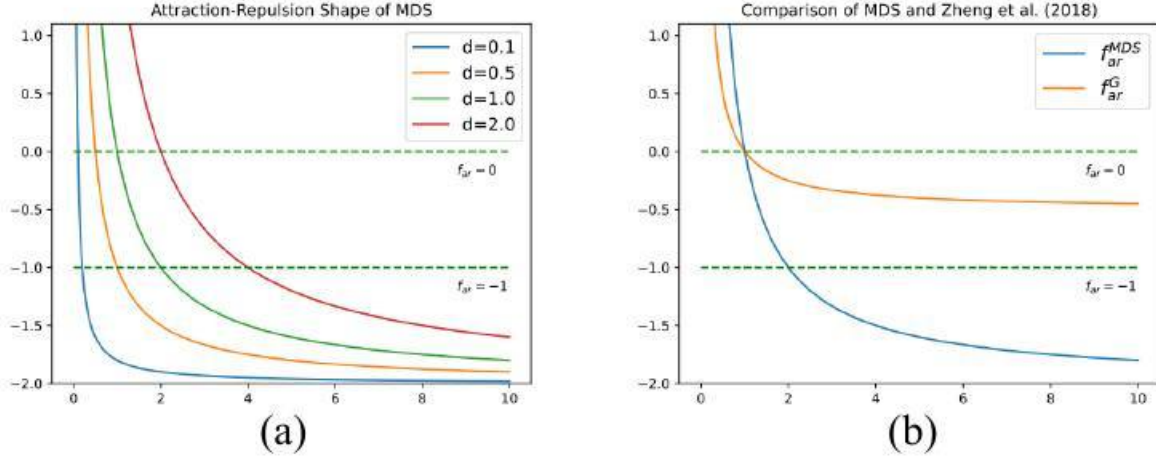


Figure 17: (a) Attraction-repulsion characteristic of MDS using f_{ar} . (b) Comparison of attraction-repulsion characteristic of MDS and the method in (Zheng et al., 2018) for $d = 1.0$.

Thus, the attraction-repulsion shape of MDS becomes

$$f_{ar}^{MDS}(\zeta) = -2 \frac{\zeta - d}{\zeta}, \quad (84)$$

where $\zeta = \|y_i - y_j\|_2$. The shape is attractive when $\zeta > d$, and in this range it is bounded within $[-2, 0]$ (Fig. 17(a)). This is ill-suited to convergence, however: only values within $[-1, 0]$ contract while the rest expand, and values below -2 flip the points and expand them. To circumvent this, (Zheng et al., 2018) adopts an ad-hoc formulation inspired by force-directed graphs:

$$f_{ar}^G(\zeta) = -\frac{1}{2} \frac{\zeta - d}{\zeta}, \quad (85)$$

which works for an effective learning rate $\lambda \leq 1$, since f_{ar}^G is bounded within $[-0.5, 0]$. For $\zeta < d$, $f_{ar}^{MDS}(> 0)$ instead repels. This attraction-repulsion interaction works best when the initialization already sits close to the desired output. Continued optimization of a pair oscillates around the distance ($\zeta = d$) where $f_{ar} = 0$, giving a notion of distance analogous to UMAP's ζ_{-1} . No learning rate drives this distance to zero—indeed, this is precisely the MDS objective—so the clusters are typically fuzzy compared to UMAP and PaCMAP (for illustrations, see (Lambert et al., 2022), (de Bodt et al., 2025), and (Kury et al., 2025)). Note that (Lambert et al., 2022) recasts the MDS loss as a quartet stress (a loss over four samples) with a relative distance (distance divided by the sum of six distances in the quartet), which enables global regularization and cuts computation relative to majorization.

H Construction of the High-Dimensional Graph

Stochastic Neighbor Embedding (SNE) (Hinton and Roweis, 2002) is the foundation of modern dimensionality reduction algorithms. It constructs a high-dimensional graph of the dataset $X = \{x_i \in \mathbb{R}^n | i = 1, \dots, N\}$ through the following system of equations:

$$w_{ij} = \frac{p_{j|i} + p_{i|j}}{2N}, \quad (86)$$

$$w_{j|i} = \frac{\exp(-\|x_i - x_j\|_2^2 / 2\sigma_i^2)}{\sum_{t \neq v} \exp(-\|x_t - x_v\|_2^2 / 2\sigma_t^2)}, \quad (87)$$

where σ_i^2 is chosen to match a user-defined value *perplexity*, P , defined as

$$P = 2^{H_i} \quad (88)$$

$$H_i = \sum_{j \neq i} w_{j|i} \log_2 w_{j|i}. \quad (89)$$

On the other hand, UMAP constructs its high-dimensional graph by the following system of equations relying on the k-nearest neighbor (k-NN) algorithm:

$$p_{i|j} = \begin{cases} \exp\left(-\frac{d(x_i, x_j) - \rho_i}{\sigma_i}\right) & \text{if } x_j \in \text{KNN}(x_i, k) \\ 0 & \text{otherwise} \end{cases}, \quad (90)$$

$$\rho_i = \min_{x_j \in \text{KNN}(x_i, k)} d(x_i, x_j), \quad (91)$$

where $\text{KNN}(x_i, k)$ is the set of k -nearest neighbors of the point x_i and σ_i is a scaling parameter such that $\sum_j p_{i|j} = \log_2(k)$. The graph is then symmetrized by a t-conorm:

$$p_{i,j} = p_{i|j} + p_{j|i} - p_{i|j}p_{j|i}. \quad (92)$$

PaCMAP uses just the k-NN graph for the affinities (all equal to 1) with a self-tuning distance measure (Zelnik-Manor and Perona 2004),

$$d_{i,j}^2 = \frac{\|x_i - x_j\|^2}{\sigma_i \sigma_j}, \quad (93)$$

where σ_i is the average distance between x_i and its Euclidean fourth-to-sixth nearest neighbors. This serves the same purpose as the corresponding σ_i in Eqs. (87) and (90), even though all three are defined differently.

Whatever the choice of high-dimensional graph, our analysis shows that the optimization—and specifically the attraction and repulsion shapes—remains the primary driver of the low-dimensional embedding.

I Detailed Results for Varying λ_a and λ_b

This section presents detailed results for the experiments in Fig. 11(h,i). We obtained these results by swapping UMAP’s attraction and repulsion shapes for those of the other methods. Figure 18 reproduces the main-text results. Figures 19-22 show the individual embeddings for each choice of λ_a and λ_r per method, together with their initialization, a reference embedding under annealed λ , and the cases where λ_a or λ_r is set to 0. For PaCMAP, which introduces mid-near pairs, we include an additional reference with those pairs. Figures 23-27 and Figures 28-32 give the corresponding results on FMNIST and single-cell transcriptomes, respectively.

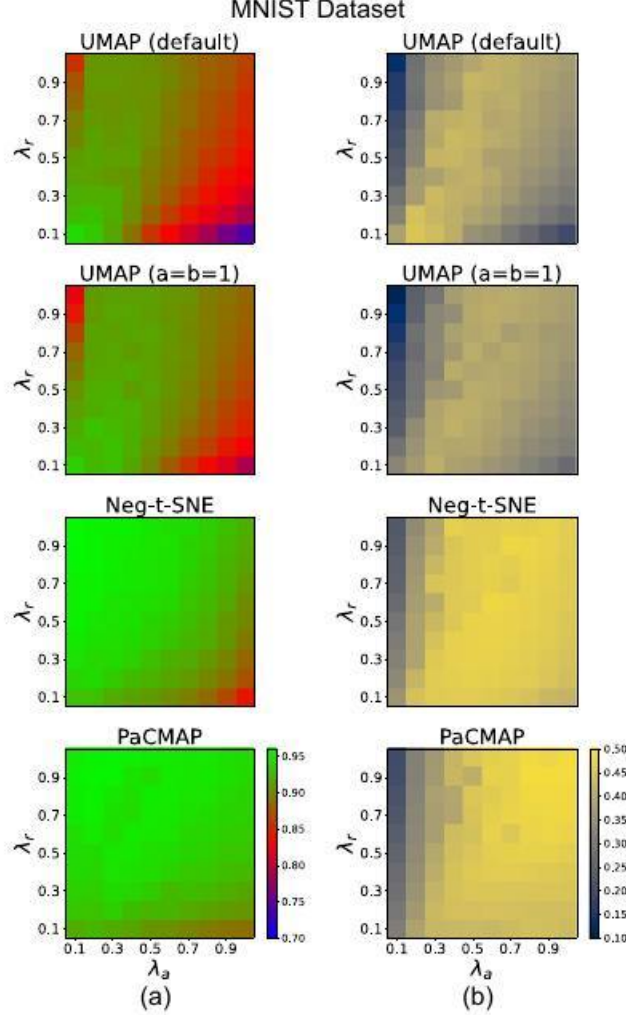


Figure 18: (a) Trustworthiness and (b) Silhouette score of different methods for the MNIST dataset.

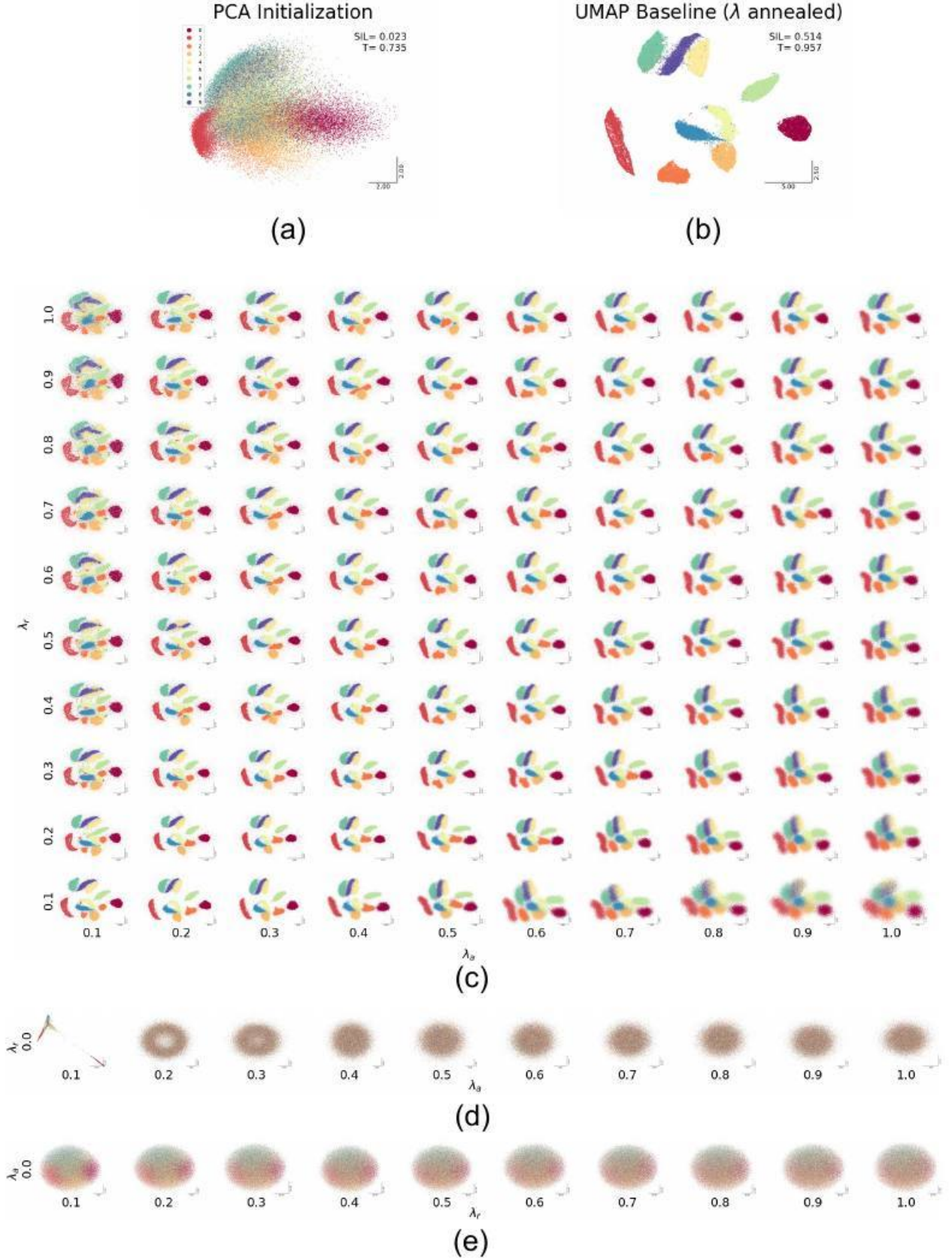


Figure 19: Varying λ_a and λ_r for the MNIST dataset using UMAP's attraction and repulsion shapes. (a) Initialization for the embeddings. (b) Baseline when λ is annealed from 1. (c) The embeddings when λ_a and λ_r vary (without any annealing). (d) When λ_r is set to 0 (no repulsion), the attractive force alone cannot produce any cluster. (e) Similarly, when λ_a is set to 0 (no attraction), the repulsive force alone cannot produce any clusters.

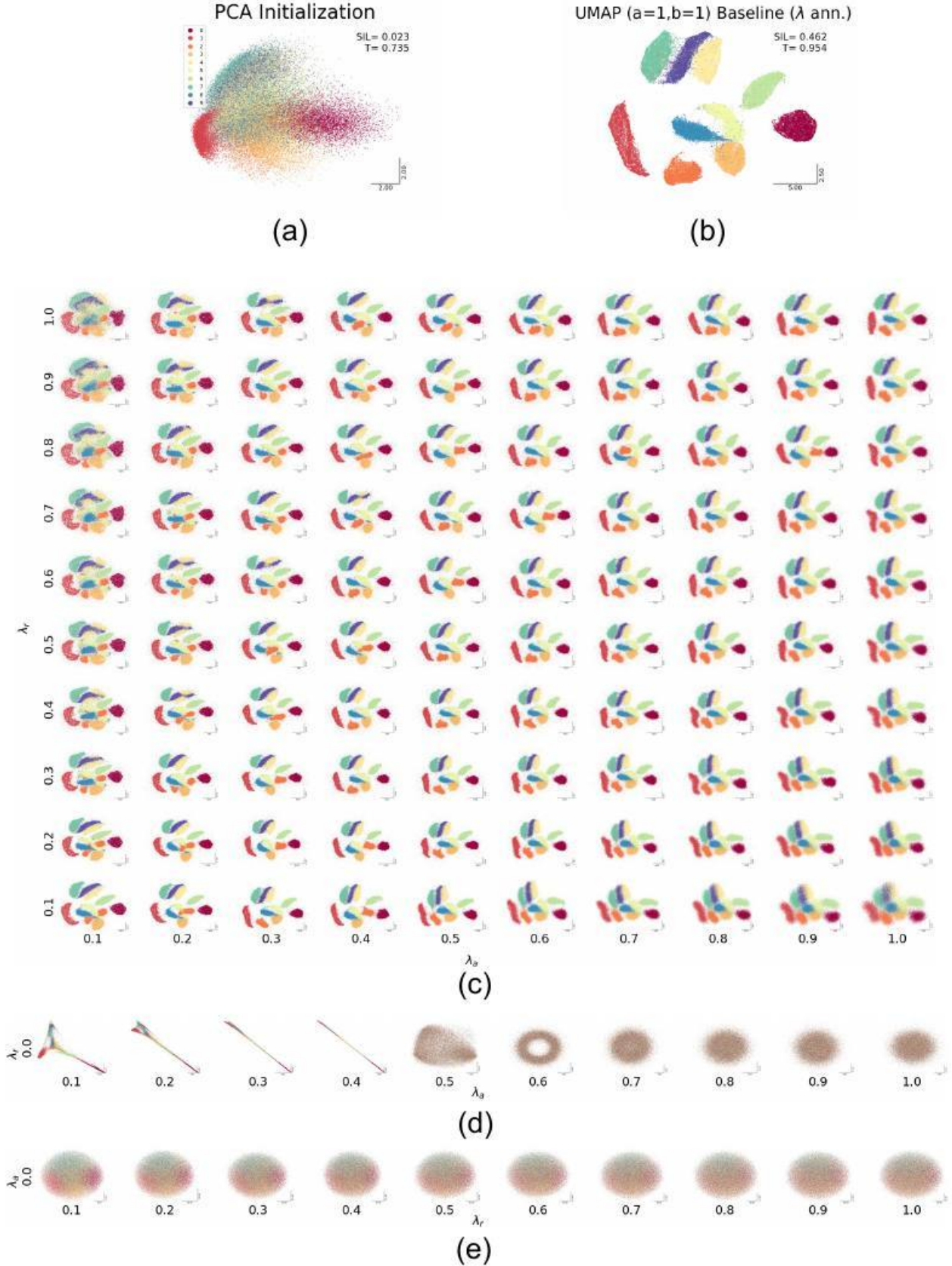


Figure 20: Varying λ_a and λ_r for the MNIST dataset using UMAP's attraction and repulsion shapes (by setting $a = 1$ and $b = 1$). (a) Initialization for the embeddings. (b) Baseline when λ is annealed from 1. (c) The embeddings when λ_a and λ_r vary (without any annealing). (d) When λ_r is set to 0 (no repulsion), the attractive force alone cannot produce any cluster. (e) Similarly, when λ_a is set to 0 (no attraction), the repulsive force alone cannot produce any clusters.

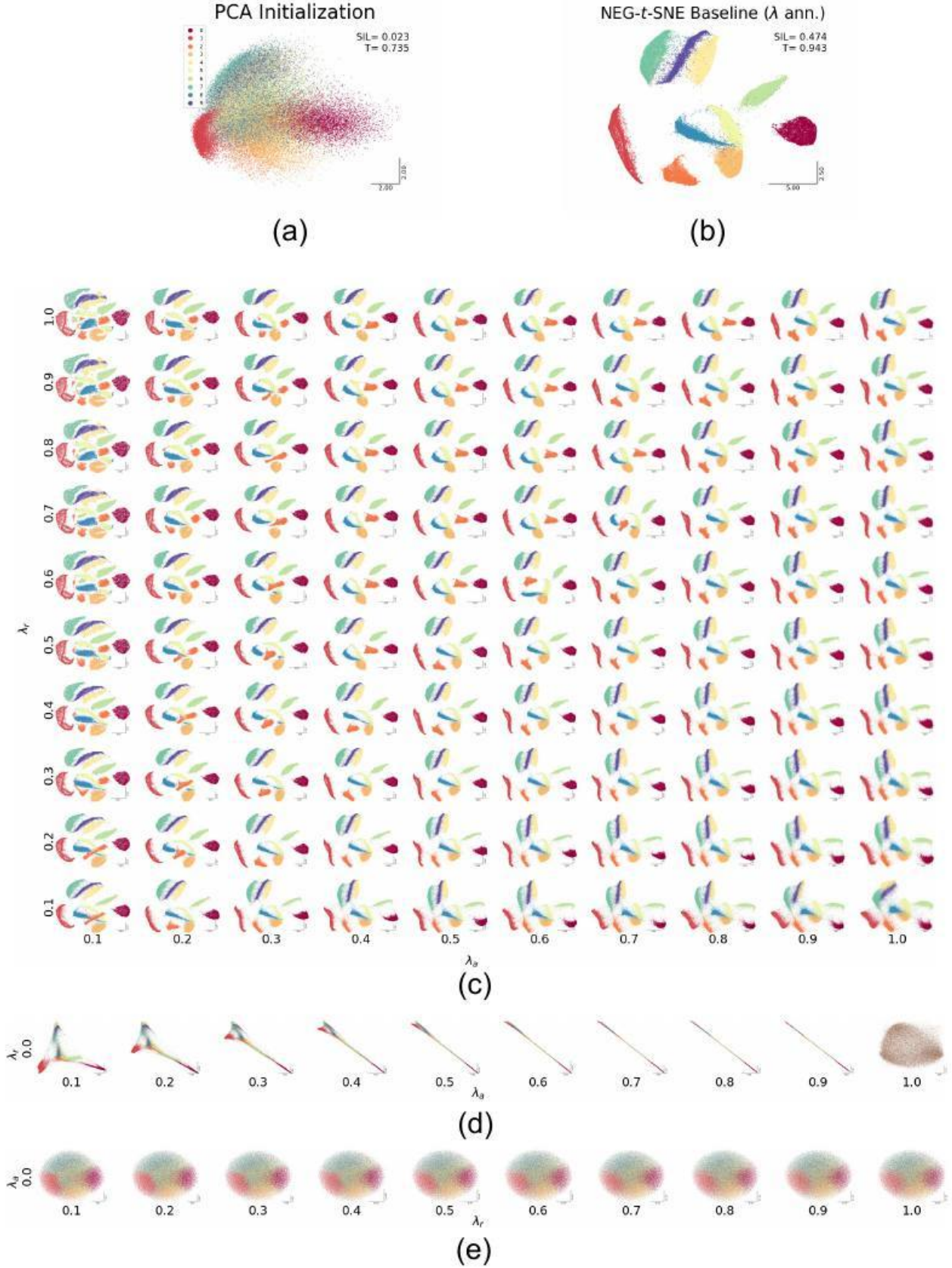


Figure 21: Varying λ_a and λ_r for the MNIST dataset using NEG-t-SNE's attraction and repulsion shapes. (a) Initialization for the embeddings. (b) Baseline when λ is annealed from 1. (c) The embeddings when λ_a and λ_r vary (without any annealing). (d) When λ_r is set to 0 (no repulsion), the attractive force alone cannot produce any cluster. (e) Similarly, when λ_a is set to 0 (no attraction), the repulsive force alone cannot produce any clusters.

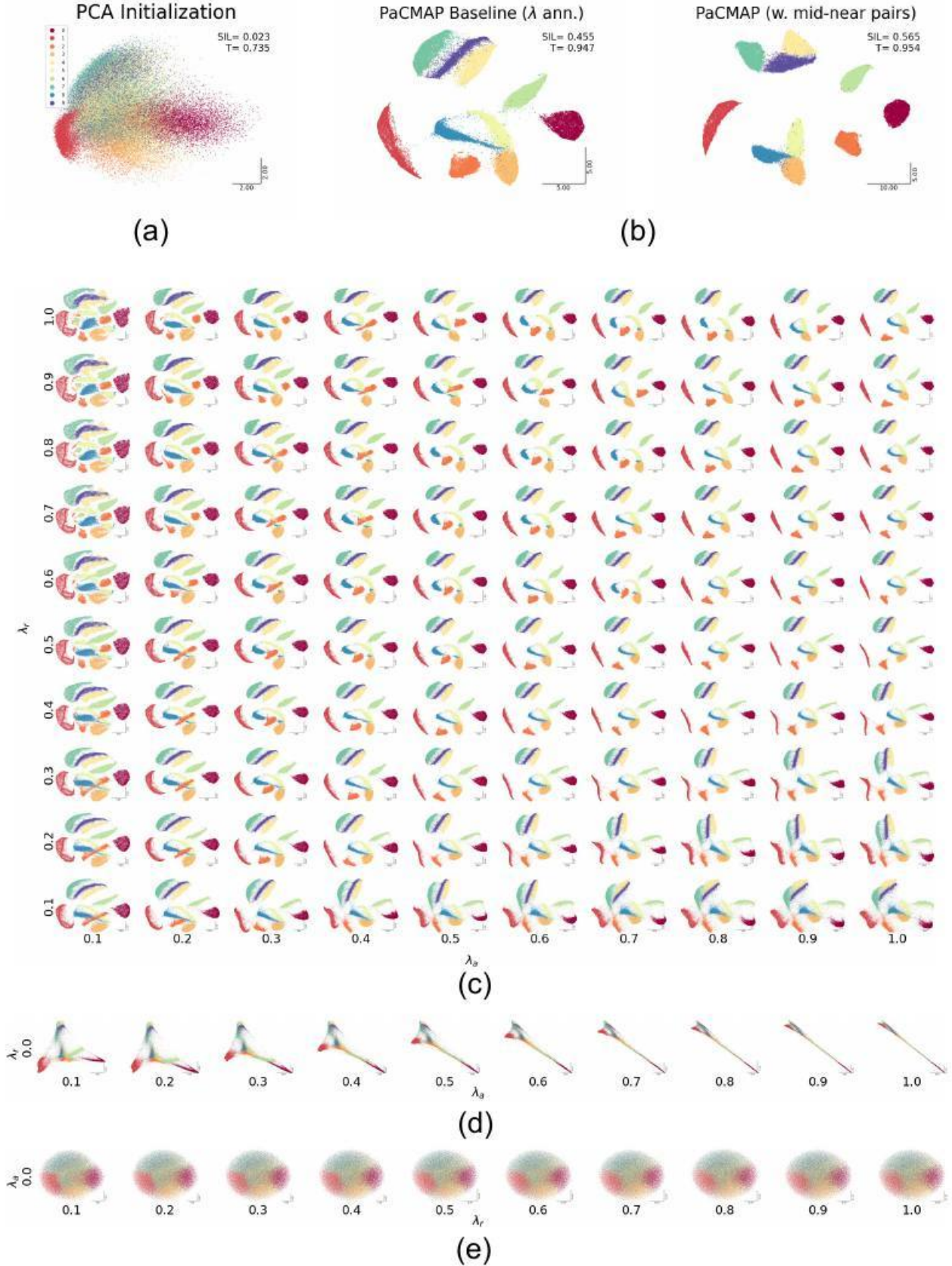


Figure 22: Varying λ_a and λ_r for the MNIST dataset using PaCMAP's attraction and repulsion shapes. (a) Initialization for the embeddings. (b) Left: Baseline when λ is annealed from 1 (mid-near points are excluded to observe the interaction of attraction-repulsion alone), right: when mid-near points are considered. (c) The embeddings when λ_a and λ_r vary (without any annealing). (d) When λ_r is set to 0 (no repulsion), the attractive force alone cannot produce any cluster. (e) Similarly, when λ_a is set to 0 (no attraction), the repulsive force alone cannot produce any clusters.

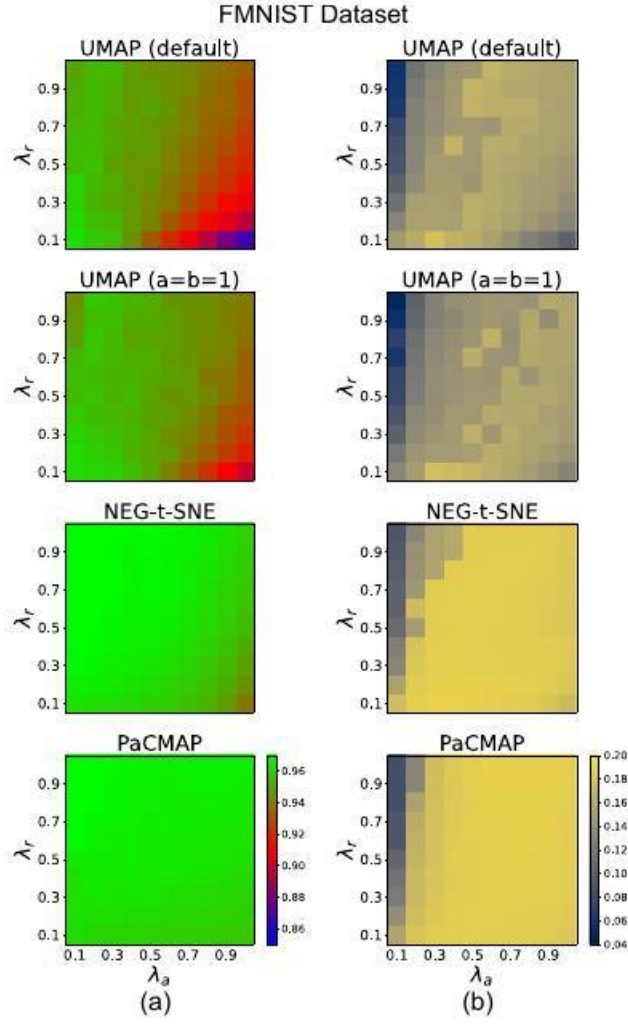


Figure 23: (a) Trustworthiness and (b) Silhouette score of different methods for the FMNIST dataset.

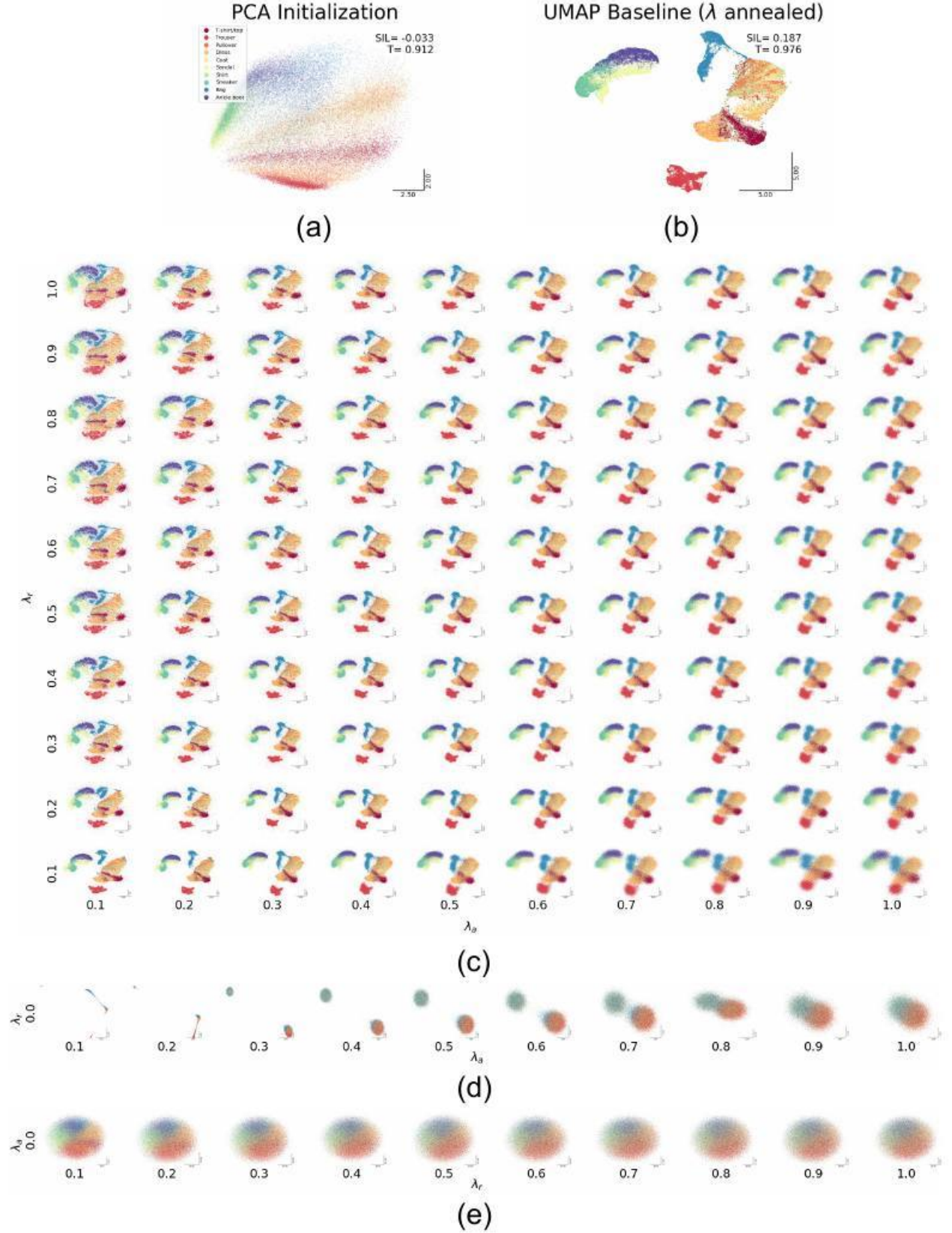


Figure 24: Varying λ_a and λ_r for the FMNIST dataset using UMAP's attraction and repulsion shapes. (a) Initialization for the embeddings. (b) Baseline when λ is annealed from 1. (c) The embeddings when λ_a and λ_r vary (without any annealing). (d) When λ_r is set to 0 (no repulsion), the attractive force alone cannot produce any cluster. (e) Similarly, when λ_a is set to 0 (no attraction), the repulsive force alone cannot produce any clusters.

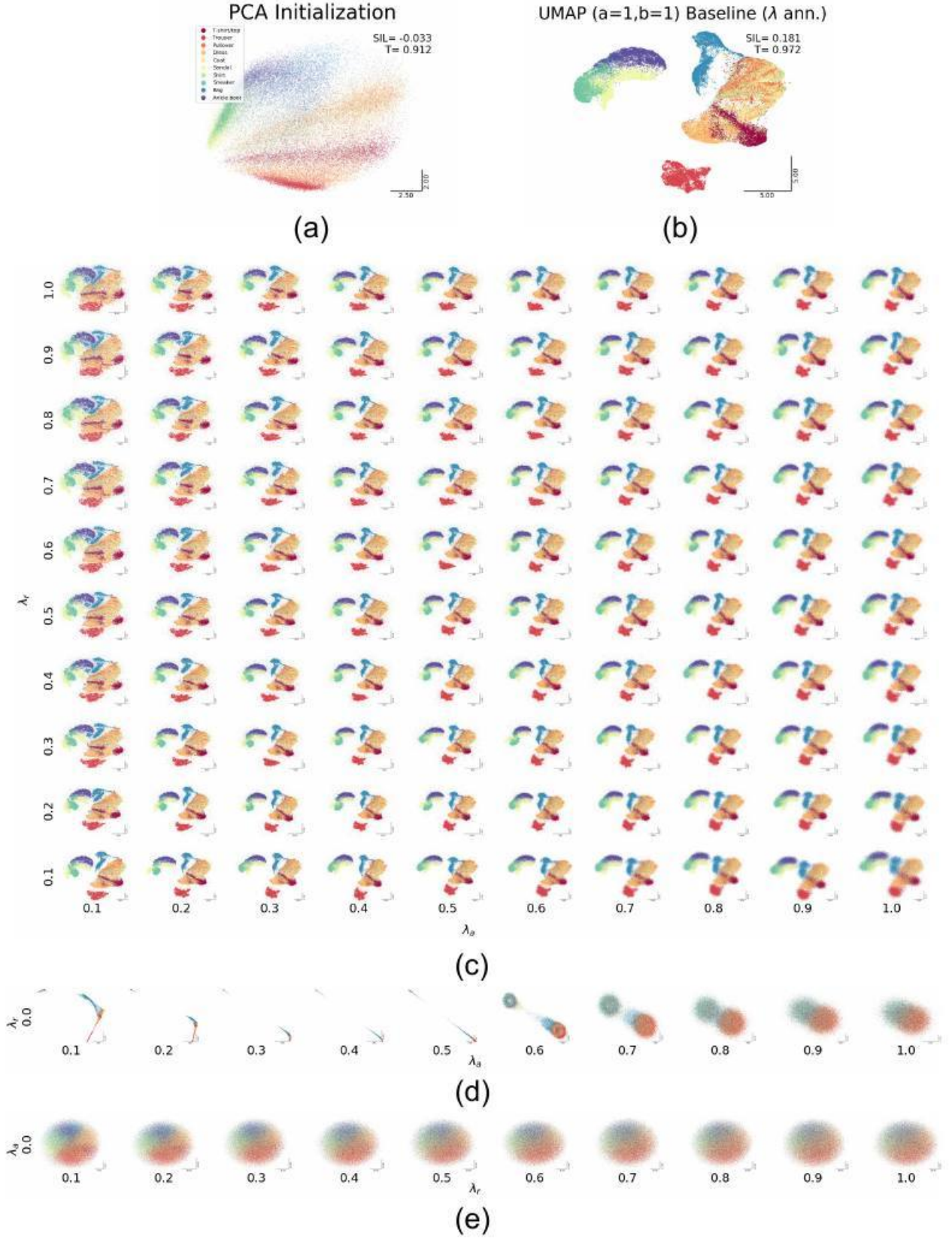


Figure 25: Varying λ_a and λ_r for the FMNIST dataset using UMAP's attraction and repulsion shapes (by setting $a = 1$ and $b = 1$). (a) Initialization for the embeddings. (b) Baseline when λ is annealed from 1. (c) The embeddings when λ_a and λ_r vary (without any annealing). (d) When λ_r is set to 0 (no repulsion), the attractive force alone cannot produce any cluster. (e) Similarly, when λ_a is set to 0 (no attraction), the repulsive force alone cannot produce any clusters.

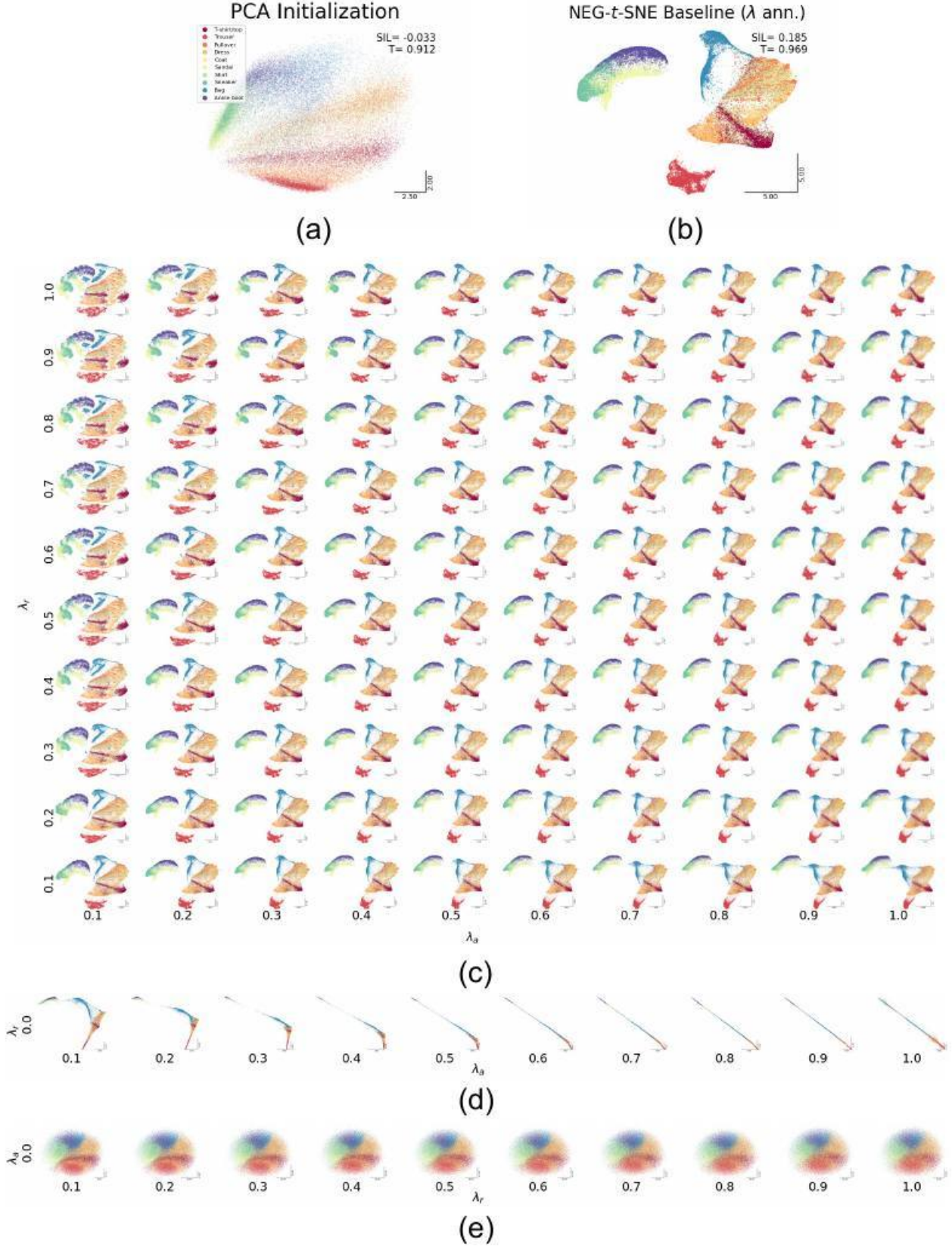


Figure 26: Varying λ_a and λ_r for the FMNIST dataset using NEG-t-SNE's attraction and repulsion shapes. (a) Initialization for the embeddings. (b) Baseline when λ is annealed from 1. (c) The embeddings when λ_a and λ_r vary (without any annealing). (d) When λ_r is set to 0 (no repulsion), the attractive force alone cannot produce any cluster. (e) Similarly, when λ_a is set to 0 (no attraction), the repulsive force alone cannot produce any clusters.

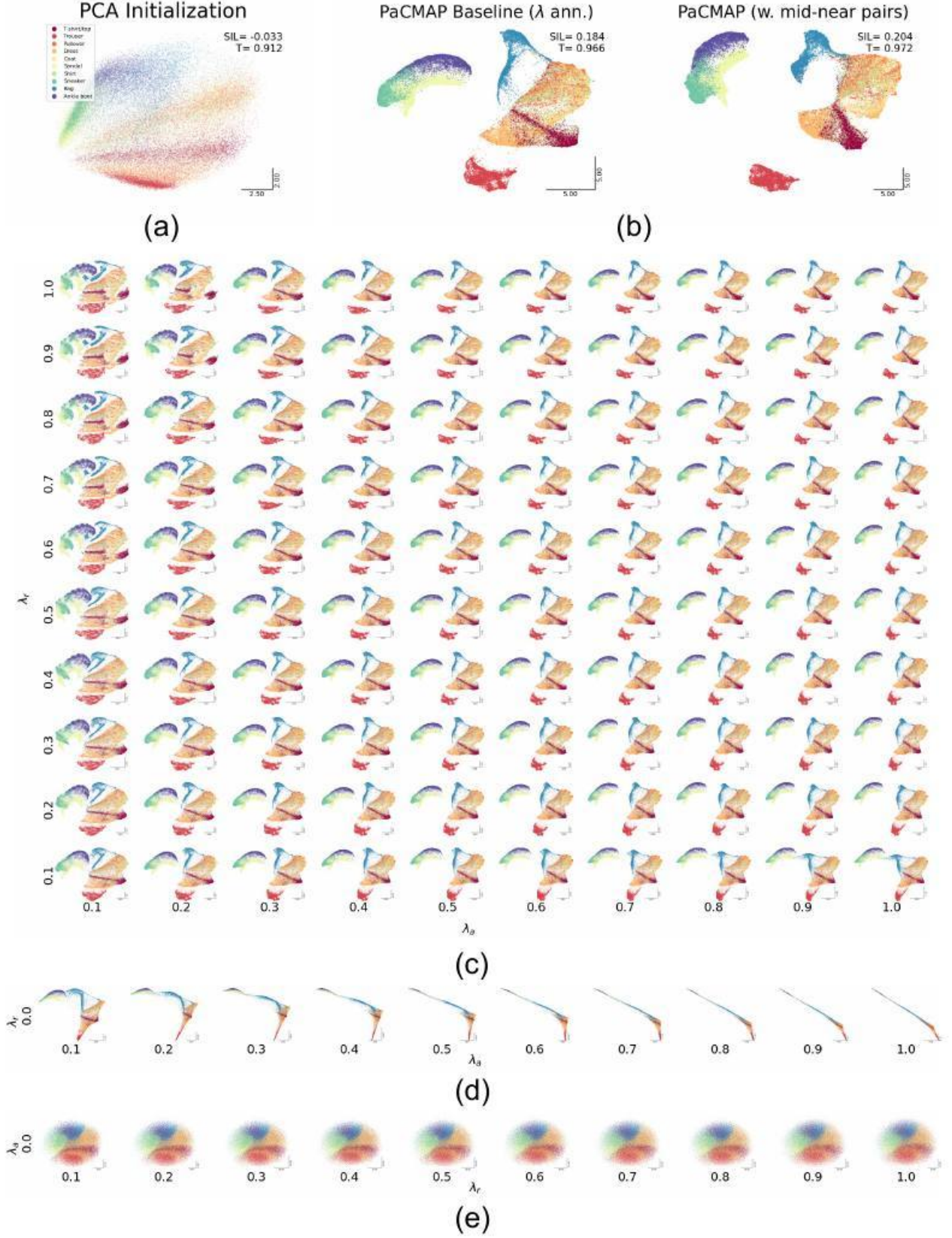


Figure 27: Varying λ_a and λ_r for the FMNIST dataset using PaCMAP's attraction and repulsion shapes. (a) Initialization for the embeddings. (b) Left: Baseline when λ is annealed from 1 (mid-near points are excluded to observe the interaction of attraction-repulsion alone), right: when mid-near points are considered. (c) The embeddings when λ_a and λ_r vary (without any annealing). (d) When λ_r is set to 0 (no repulsion), the attractive force alone cannot produce any cluster. (e) Similarly, when λ_a is set to 0 (no attraction), the repulsive force alone cannot produce any clusters.

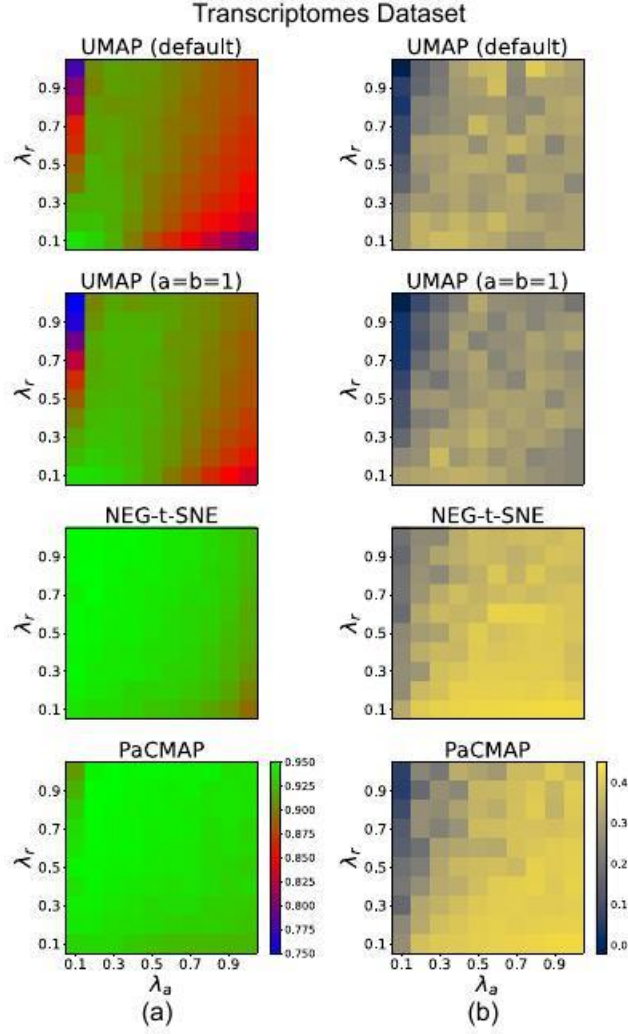


Figure 28: (a) Trustworthiness and (b) Silhouette score of different methods for the Single-cell transcriptomes dataset.

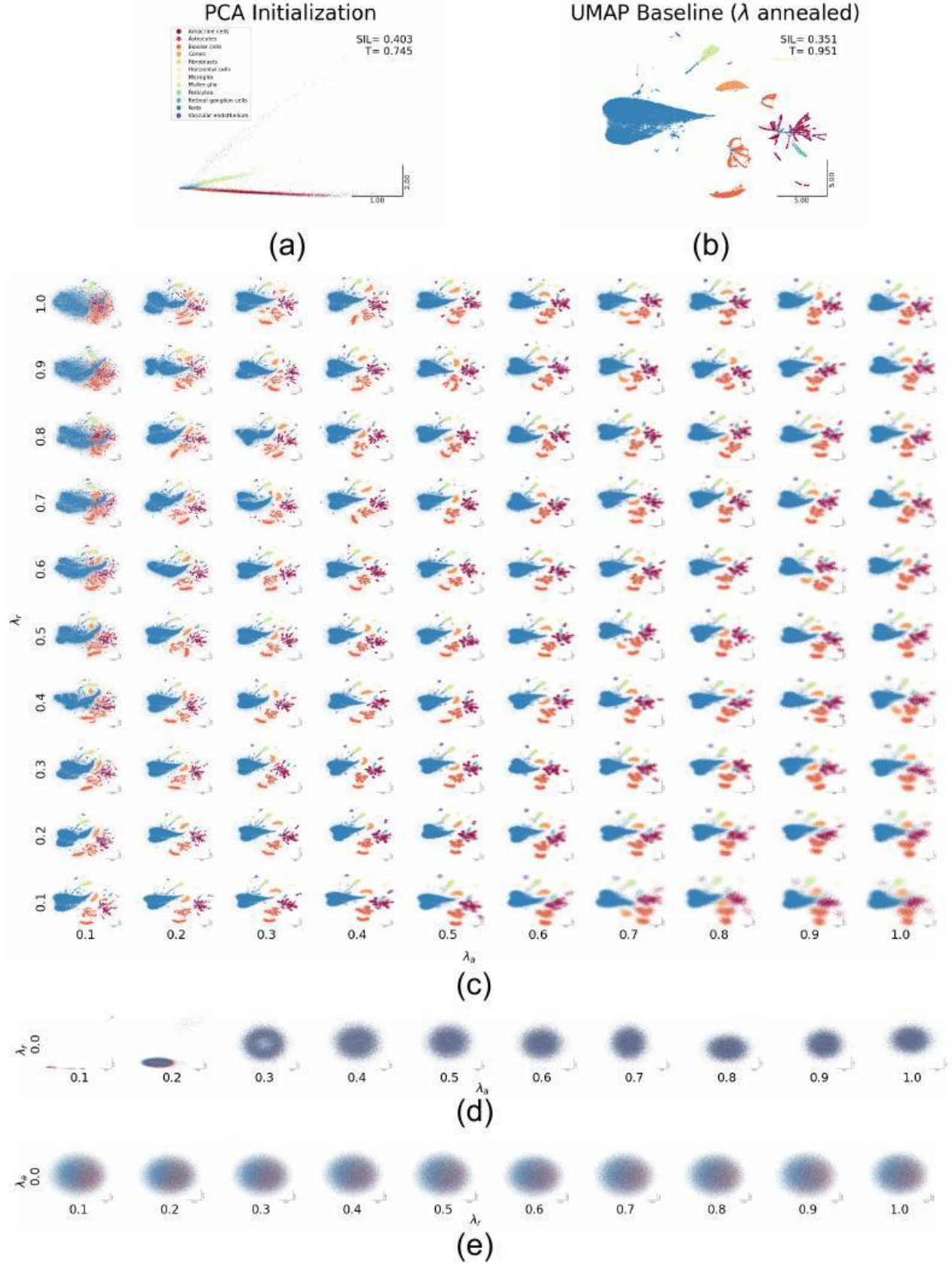


Figure 29: Varying λ_a and λ_r for the single-cell transcriptomes dataset using UMAP's attraction and repulsion shapes. (a) Initialization for the embeddings. (b) Baseline when λ is annealed from 1. (c) The embeddings when λ_a and λ_r vary (without any annealing). (d) When λ_r is set to 0 (no repulsion), the attractive force alone cannot produce any cluster. (e) Similarly, when λ_a is set to 0 (no attraction), the repulsive force alone cannot produce any clusters.

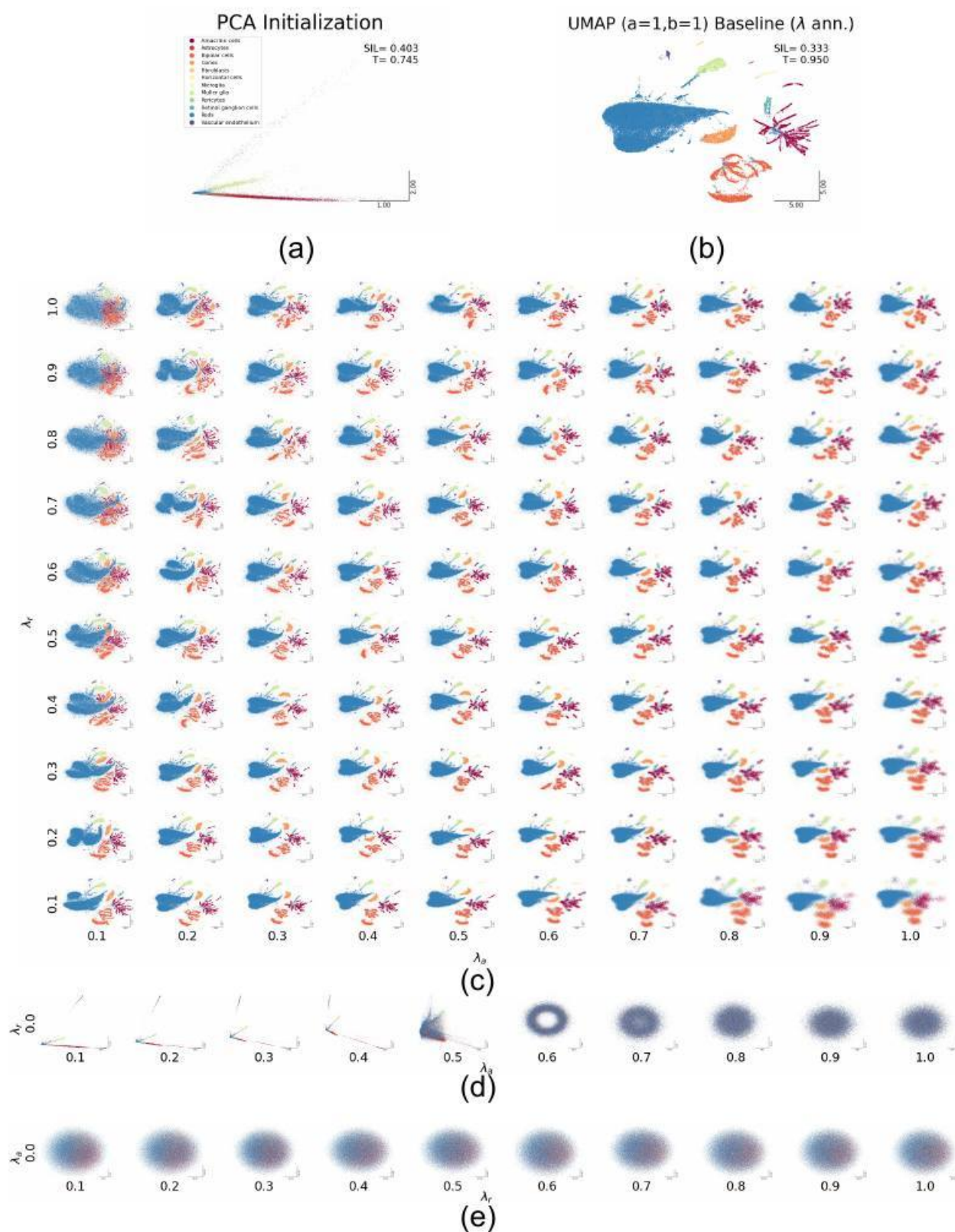


Figure 30: Varying λ_a and λ_r for the single-cell transcriptomes dataset using UMAP’s attraction and repulsion shapes (by setting $a = 1$ and $b = 1$). (a) Initialization for the embeddings. (b) Baseline when λ is annealed from 1. (c) The embeddings when λ_a and λ_r vary (without any annealing). (d) When λ_r is set to 0 (no repulsion), attractive force alone cannot produce any cluster. (e) Similarly, when λ_a is set to 0 (no attraction), repulsive force alone cannot produce any clusters.

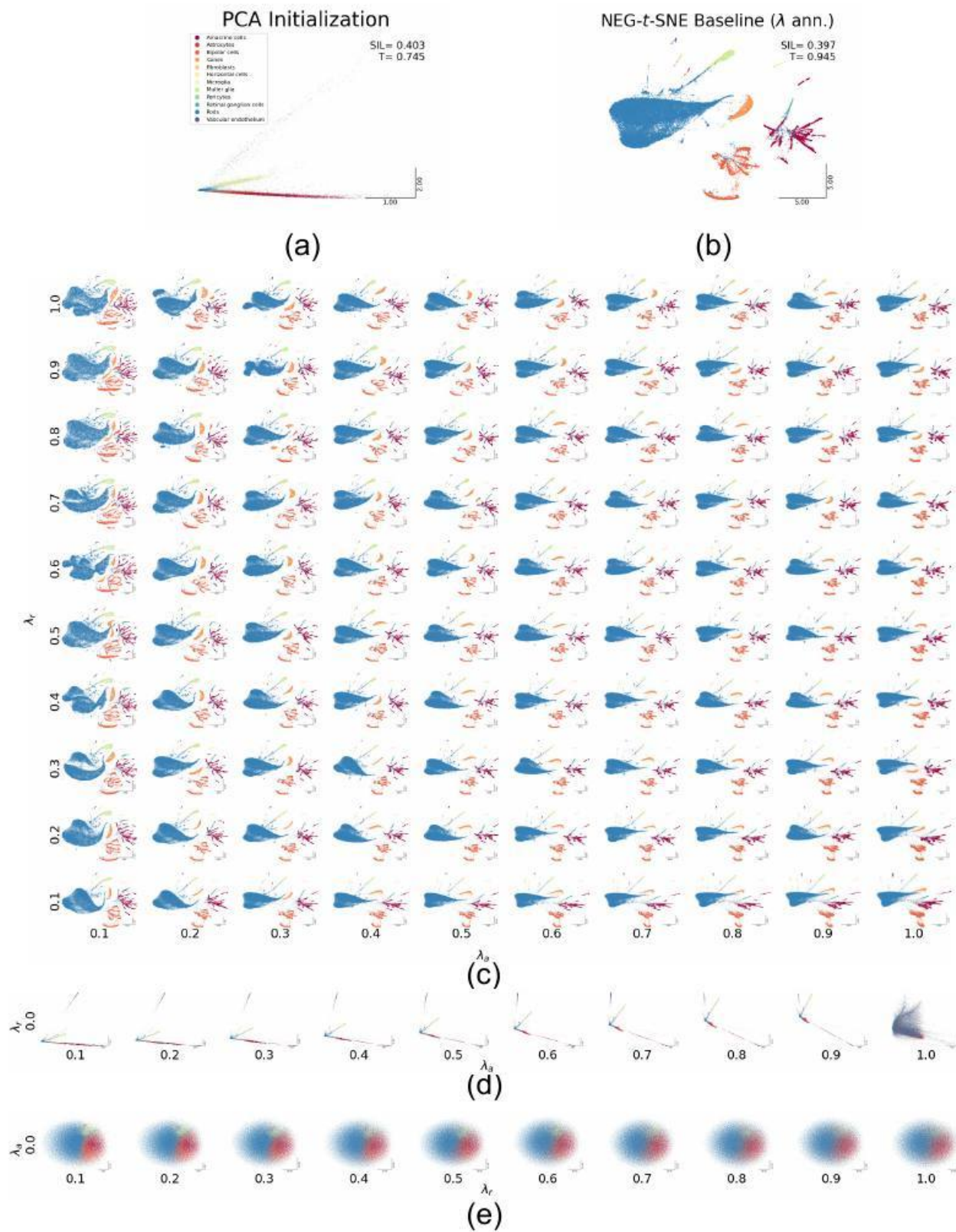


Figure 31: Varying λ_a and λ_r for the single-cell transcriptomes dataset using NEG-*t*-SNE’s attraction and repulsion shapes. (a) Initialization for the embeddings. (b) Baseline when λ is annealed from 1. (c) The embeddings when λ_a and λ_r vary (without any annealing). (d) When λ_r is set to 0 (no repulsion), the attractive force alone cannot produce distinct clusters. (e) Similarly, when λ_a is set to 0 (no attraction), the repulsive force alone cannot produce any clusters.

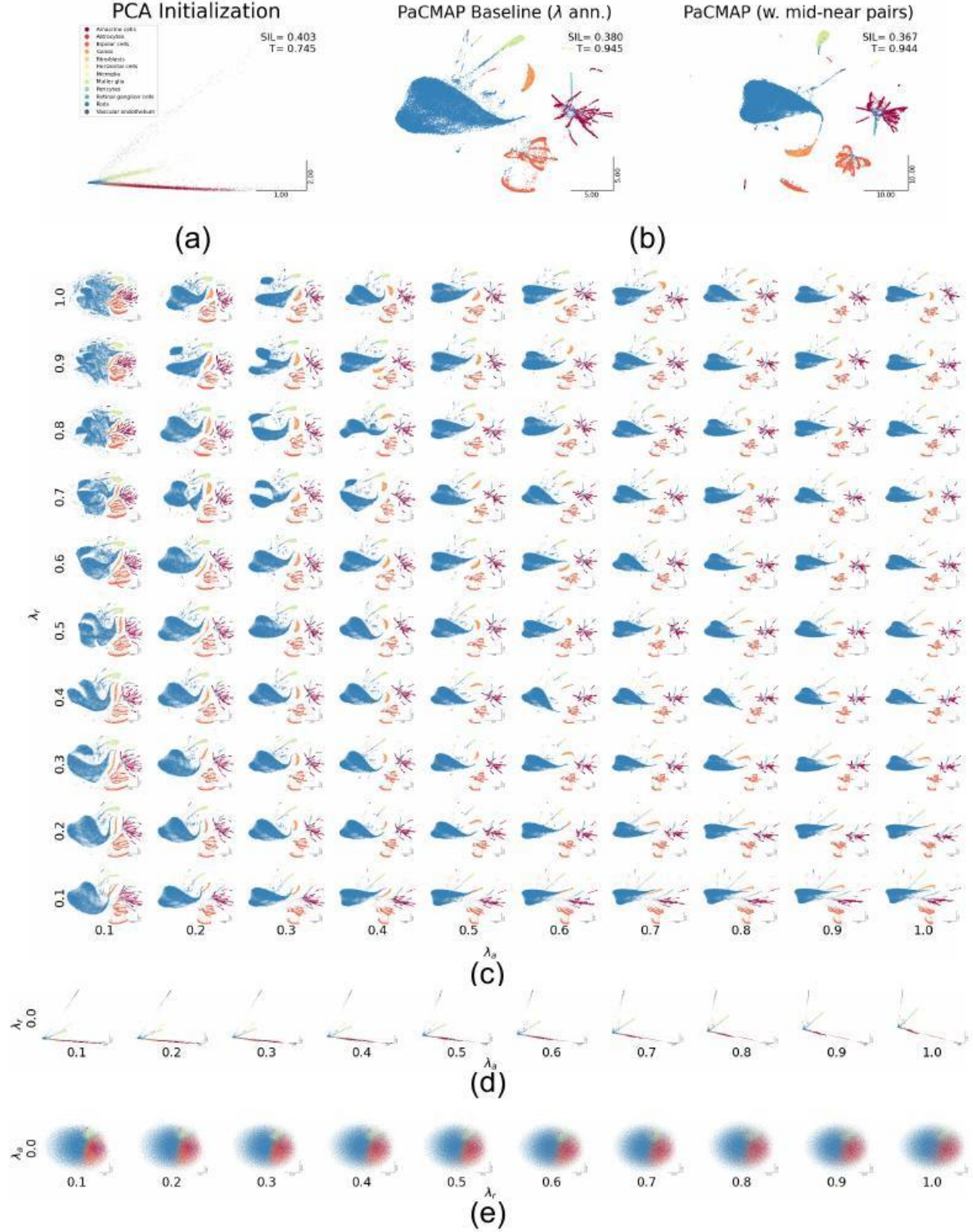


Figure 32: Varying λ_a and λ_r for the single-cell transcriptomes dataset using PaCMAP's attraction and repulsion shapes. (a) Initialization for the embeddings. (b) Left: Baseline when λ is annealed from 1 (mid-near points are excluded to observe the interaction of attraction-repulsion alone), right: when mid-near points are considered. (c) The embeddings when λ_a and λ_r vary (without any annealing). (d) When λ_r is set to 0 (no repulsion), the attractive force alone cannot produce any cluster. (e) Similarly, when λ_a is set to 0 (no attraction), the repulsive force alone cannot produce any clusters.

J Mix and Match Approach

Throughout our analysis we treated the attraction and repulsion shapes as independent. This independence has a direct consequence: we can freely combine attraction and repulsion shapes drawn from different loss functions and principles. Figure 33 shows several such combinations. We restricted the attraction shapes to $[-1, 0]$ (to satisfy Proposition 4.1) so they become insensitive to the learning rate, and we used finite repulsion shapes to suppress the effect of large repulsive forces at small distances. Holding the learning rate constant ($\lambda = 1$) and using UMAP’s optimization throughout ensures that any change stems from the shapes alone rather than from learning-rate scheduling or the optimizer. The outcome is clear-cut: UMAP’s attraction shape (via $f_a = 0.5f_a^U$) produces better clusters (higher silhouette score), while PaCMAP’s attraction shape delivers better local structure (higher trustworthiness), the latter because PaCMAP’s attraction shape lies within $[-0.5, 0]$ and thus never flips during the updates. Confining UMAP’s and NEG- t -SNE’s attraction shapes to $[-0.5, 0]$ (via $f_a = 0.25f_a^U$ and $f_a = 0.5f_a^N$, respectively) improves local structure at the cost of clustering performance (Fig. 34).

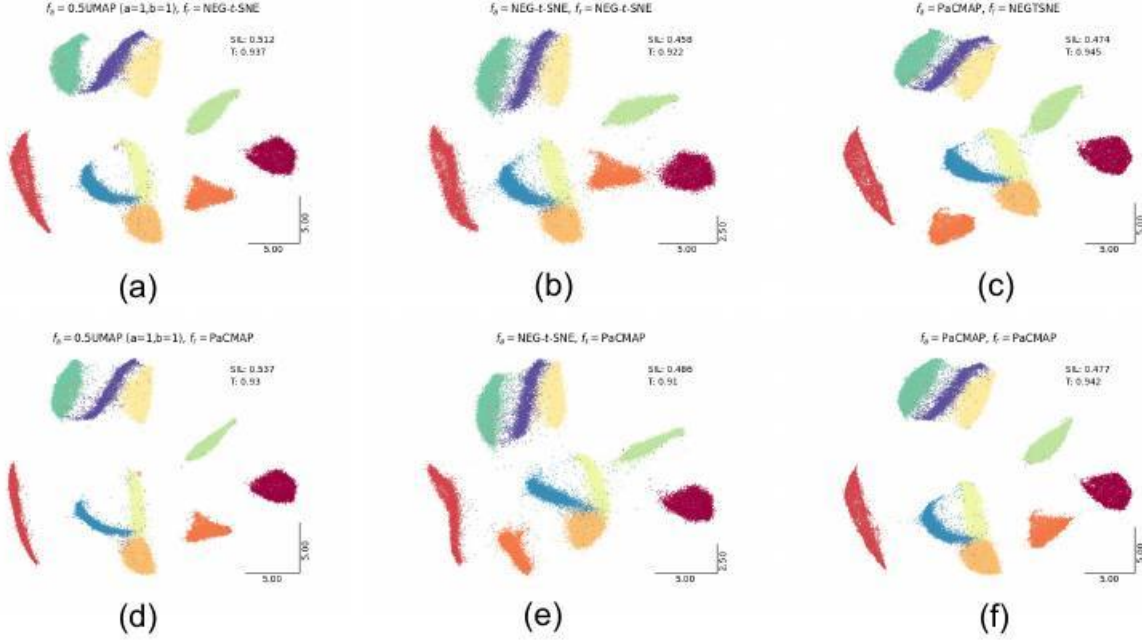


Figure 33: Embedding of MNIST by mixing and matching attraction and repulsion shapes with a constant learning rate ($\lambda = 1$). We used attraction shapes that follow Proposition 4.1 and repulsion shapes that are finite. Top row: UMAP, NEG- t -SNE, and PaCMAP’s attraction shapes with NEG- t -SNE’s repulsion shape. Bottom row: UMAP, NEG- t -SNE, and PaCMAP’s attraction shapes with PaCMAP’s repulsion shape.

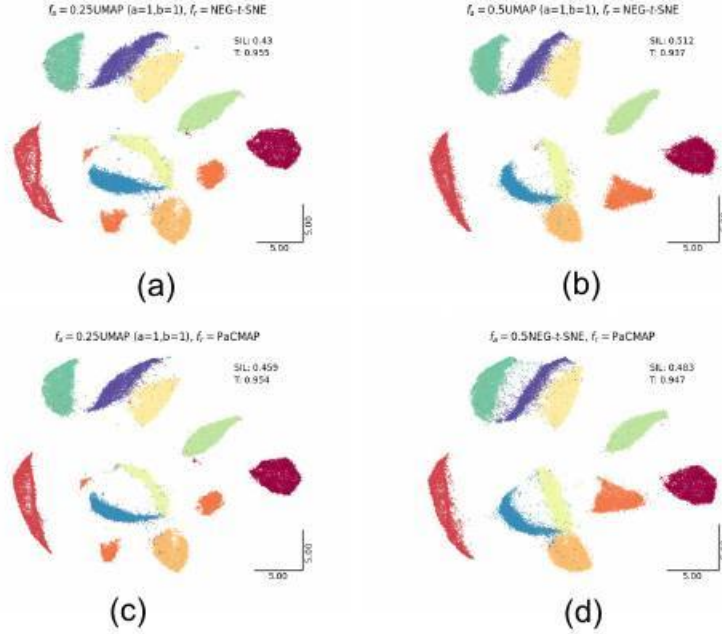


Figure 34: Embedding of MNIST by mixing and matching attraction and repulsion shapes with a constant learning rate ($\lambda = 1$). We kept the attraction shape confined to $[-0.5, 0]$ to prevent any flips during the attractive update. Top row: UMAP and NEG- t -SNE’s attraction shapes with NEG- t -SNE’s repulsion shape. Bottom row: UMAP and NEG- t -SNE’s attraction shapes with PaCMAP’s repulsion shape.

K Implementation Details

For our analysis, we implemented UMAP from scratch. We used `numba` (Lam et al., 2015) to compute an exact—rather than approximate—nearest neighbor graph with $k = 15$, and `scikit-learn`'s (Pedregosa et al., 2011) PCA implementation for PCA initialization. This implementation produced and quantified the embeddings in Figs. 1, 2, 3, 4, 6, 8, 9, and 10, and we reused it when swapping the attraction and repulsion shapes for those of the alternative methods (Figs. 18, 34, unless otherwise stated). Trustworthiness and silhouette scores were computed with the corresponding `scikit-learn` functions.

The mappings in Figs. 2, 3, 9, 10, and 15 are rotated to a reference embedding (panel (a) of each respective figure). We did this via Procrustes alignment: normalizing the embeddings (zero mean, unit norm) and then extracting rotation and scaling parameters with SciPy's (Virtanen et al., 2020) `orthogonal_procrustes` method.

To compare with Neg- t -SNE in Fig. 11, we used the original implementation of the contrastive embedding framework for both the UMAP and the Neg- t -SNE algorithms; the implementation details will be released upon acceptance.

PaCMAP and LocalMAP embeddings in Figs. 15, 16, 22(b) (right), 27(b) (right), and 32(b) (right) were obtained using the official PaCMAP package; the implementation details will be released upon acceptance.

The codes for displaying the embeddings of the PubMed dataset in Fig. 7 will be released upon acceptance.